Immunity and Immune system

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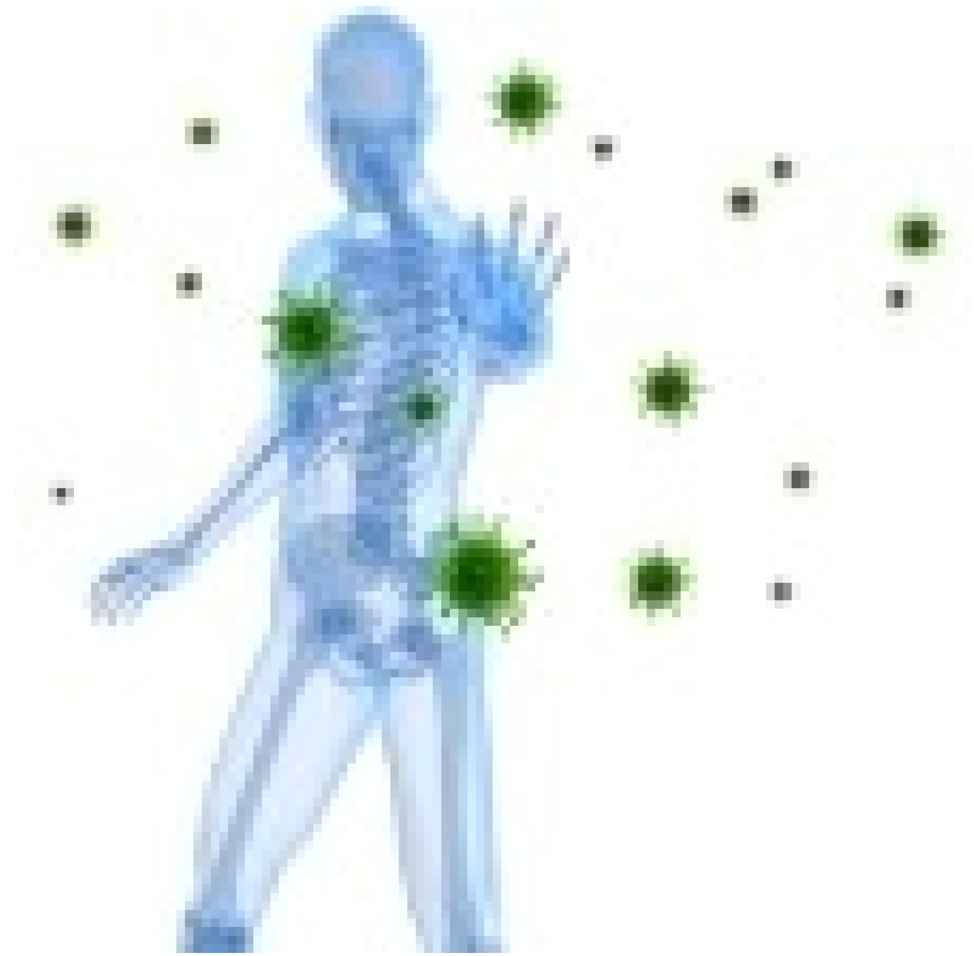


আমাদের শরীরের ইমিউন সিস্টেম সব সময় রোগজীবাণুর বিরুদ্ধে যুদ্ধ করে। কিন্তু মাঝে মাঝে শরীরের প্রতিরক্ষা ব্যবস্থা ভুল করে নিজের কোষকেই শত্রু ভেবে আক্রমণ করে— আর তখনই দেখা দেয় Autoimmune disease, যেমন: ডায়াবেটিস টাইপ–১, রিউমাটয়েড আর্থ্রাইটিস, মাল্টিপল স্ক্লেরোসিস ইত্যাদি। কাজেই এসব রোগ থেকে বাঁচতে হলে ইমিউন সিস্টেমকে নিজের দেহকোষ আক্রমণ করা থেকে বিরত রাখতে হবে। আমাদের শরীরের ইমিউন সিস্টেম দেহকোষ বা টিস্যুকে আক্রমণ করা থেকে বিরত রাখার সেই গুরুত্বপূর্ণ প্রক্রিয়ার রহস্য উদঘাটন করেছেন এই বিজ্ঞানীরা। তারা শরীরের এমন কিছু বিশেষ কোষ খুঁজে বের করেছেন, যেগুলো ইমিউন সিস্টেমকে নিজ দেহের কোষ আক্রমণ করতে বাধা দেয়। এই কোষগুলোকে বলা হয় Regulatory T-cell (T-reg)।

তাদের মূল অবদান ছিল T-reg শনাক্ত করা এবং এর কার্যকারিতা ব্যাখ্যা করা। শিমন সাকাগুচি প্রথম প্রমাণ করেন যে T-reg কোষ ইমিউন সিস্টেমকে এমনভাবে নিয়ন্ত্রণ করে যেন তা নিজ দেহকোষকে আক্রমণ না করে।

মেরি ব্রাংকাও ও ফ্রেড র‍্যামসডেল আবিষ্কার করেন একটি গুরুত্বপূর্ণ জিন — FOXP3, যা এই T-reg সেল তৈরি ও সঠিকভাবে কাজ করার জন্য অপরিহার্য। এই জিনে ত্রুটি দেখা দিলে শরীরের প্রতিরোধ ব্যবস্থা ভারসাম্য হারায় এবং বিভিন্ন অটোইমিউন রোগ দেখা দেয়।

এই আবিষ্কার চিকিৎসা বিজ্ঞানে নতুন দিগন্ত উন্মোচন করেছে, যেমন—

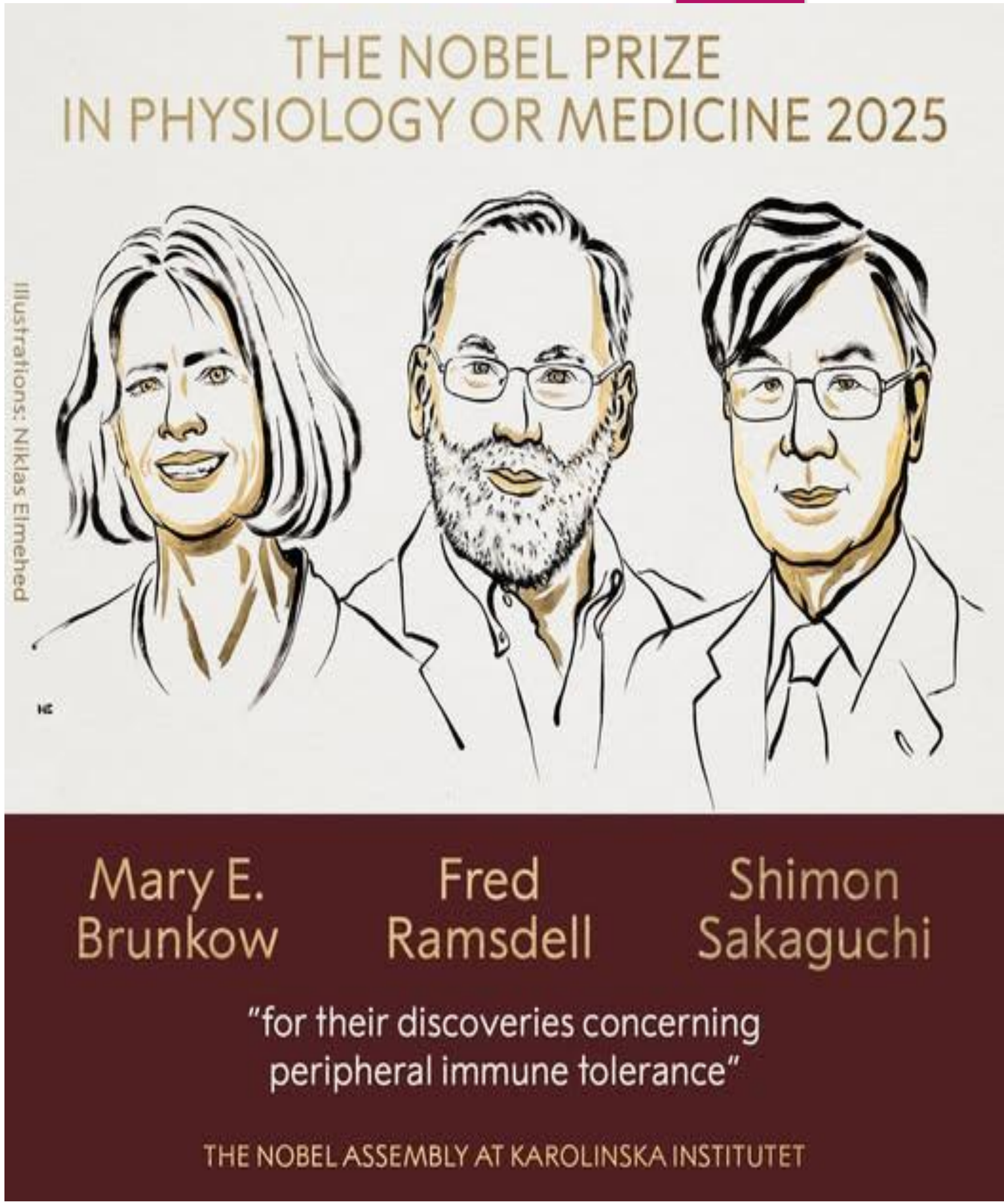
১। অটোইমিউন রোগের চিকিৎসা: T-reg কোষ ও FOXP3 জিনকে কাজে লাগিয়ে এমন ওষুধ তৈরি করা, যা ইমিউন সিস্টেমের ভারসাম্য ফিরিয়ে আনবে।

২। অঙ্গ প্রতিস্থাপনে সহায়তা: নতুন অঙ্গ/কোষ গ্রহণের সময় শরীর যেন তা প্রত্যাখ্যান না করে, সেই সহনশীলতা বাড়াতে এই জ্ঞান ব্যবহার করা।

৩। ক্যান্সার চিকিৎসায় সহায়তা: ক্যান্সার কোষ প্রায়ই রোগ প্রতিরোধ ব্যবস্থাকে ফাঁকি দিয়ে নিজেদের আড়াল করে ফেলে। T-reg কোষকে কাজে লাগিয়ে ক্যান্সার ইমিউনোথেরাপি আরও কার্যকর করা।

এই গবেষণা আমাদের শেখায়, মানবদেহের প্রতিরোধ ব্যবস্থা শুধু বহিঃশত্রুর বিরুদ্ধেই যুদ্ধ করে না, বরং তা “নিজের শত্রুকে চিনে নেয়ার” ক্ষমতাও রাখে। নোবেলজয়ী এই আবিষ্কার ভবিষ্যতে এমন অনেক কঠিন রোগের চিকিৎসা সহজ করবে, যেখানে শরীরের নিজের কোষই শত্রু হিসেবে হিসেবে কাজ করে।

Written By, Professor Anowarul Azim Akhand, DU GEB, VC MBSTU, Tangail.



What is immune System?

Even though we are constantly bombarded by microorganisms, we rarely become sick. Why?

- In Simple, it is the study of our **body's defense mechanisms** against infections and disease.
- The immune system is a complex network **of organs, cells and proteins** that defends the body ***against infection***, whilst protecting the body's own cells.
- The immune system **keeps a record of every antigens** (pathogen/germ/foreign/harmful substances) it has ever defeated so it can recognize and destroy the antigen quickly if it enters the body again.

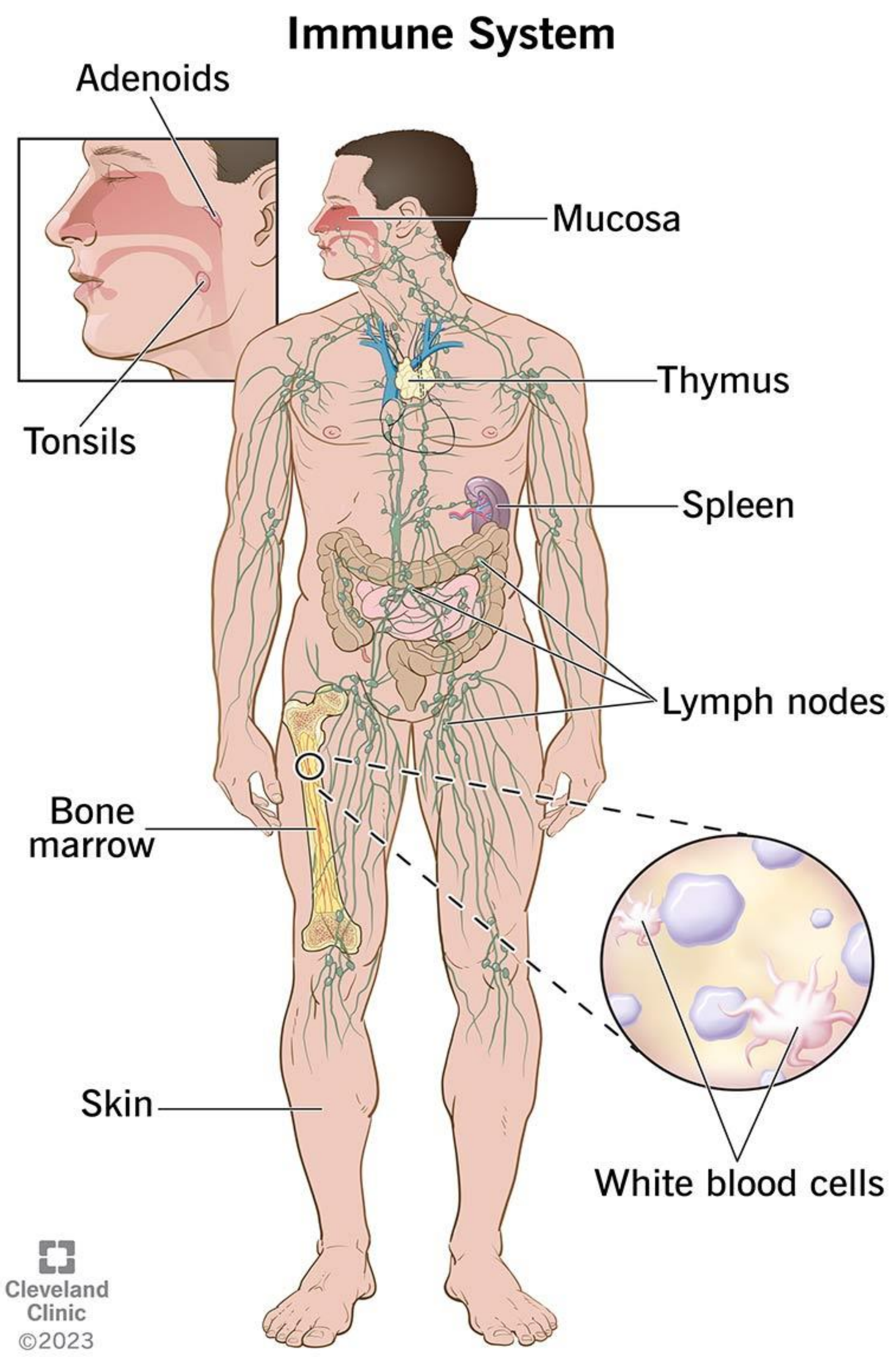
Components of immune System?

The immune system is made up of special organs, cells and chemicals that fight infections (microbes) and Cancers.

<u>The main parts of the immune system are:</u>

- The thymus and
- The spleen
- Lymph Nodes
- The bone marrow
- Adenoids, Tonsils
- The complement system
- The lymphatic system
- White blood cells
- Antibodies

These are the parts of your immune system that actively fight infection



Mechanisms of immune system and their cooperation

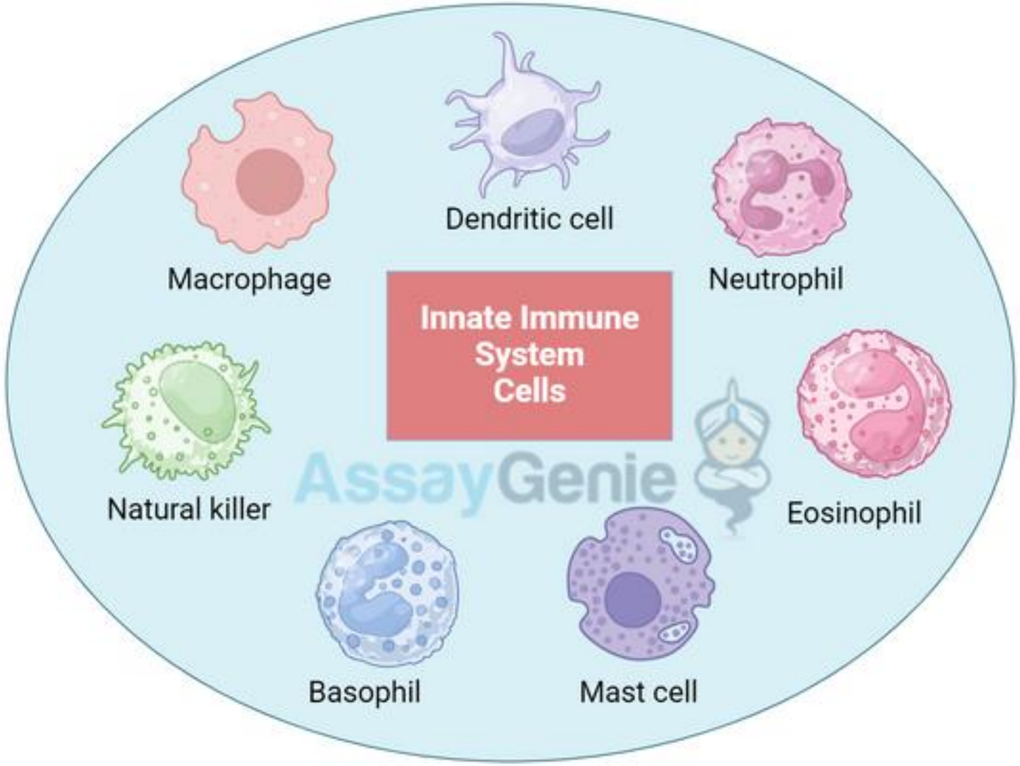
1/ Innate (non-specific) immune system:

The innate immune system is the first line of defense against pathogens, providing a rapid response to infections. This system comprises various cell types, each with specialized roles in detecting and eliminating invaders.

Immunity that is present at birth and lasts a person's entire life. Innate immunity is the first response of the body's immune system to a harmful foreign substance. When foreign substances, such as bacteria or viruses, enter the body, certain cells in the immune system can quickly respond and try to destroy them. Innate immunity also includes barriers, such as skin, mucous membranes, tears, and stomach acid, that help keep harmful substances from entering the body

- innate, not developed after the exposition to infection
- uniform response, prompt, no immunological memory
- mechanical barriers (mucosa, skin)
- phagocytic cells (microphages, macrophages)
- acute phase proteins (CRP)
- complement system

https://www.youtube.com/watch?v=sNIcR1PAGIk



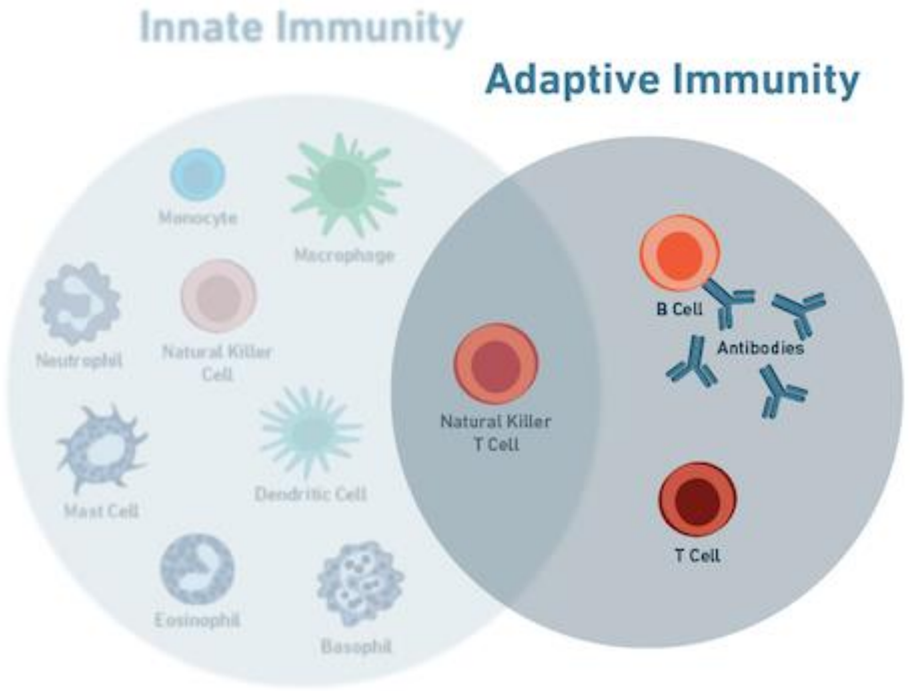
Mechanisms of immune system and their cooperation

2/ Adaptive (specific) immune system

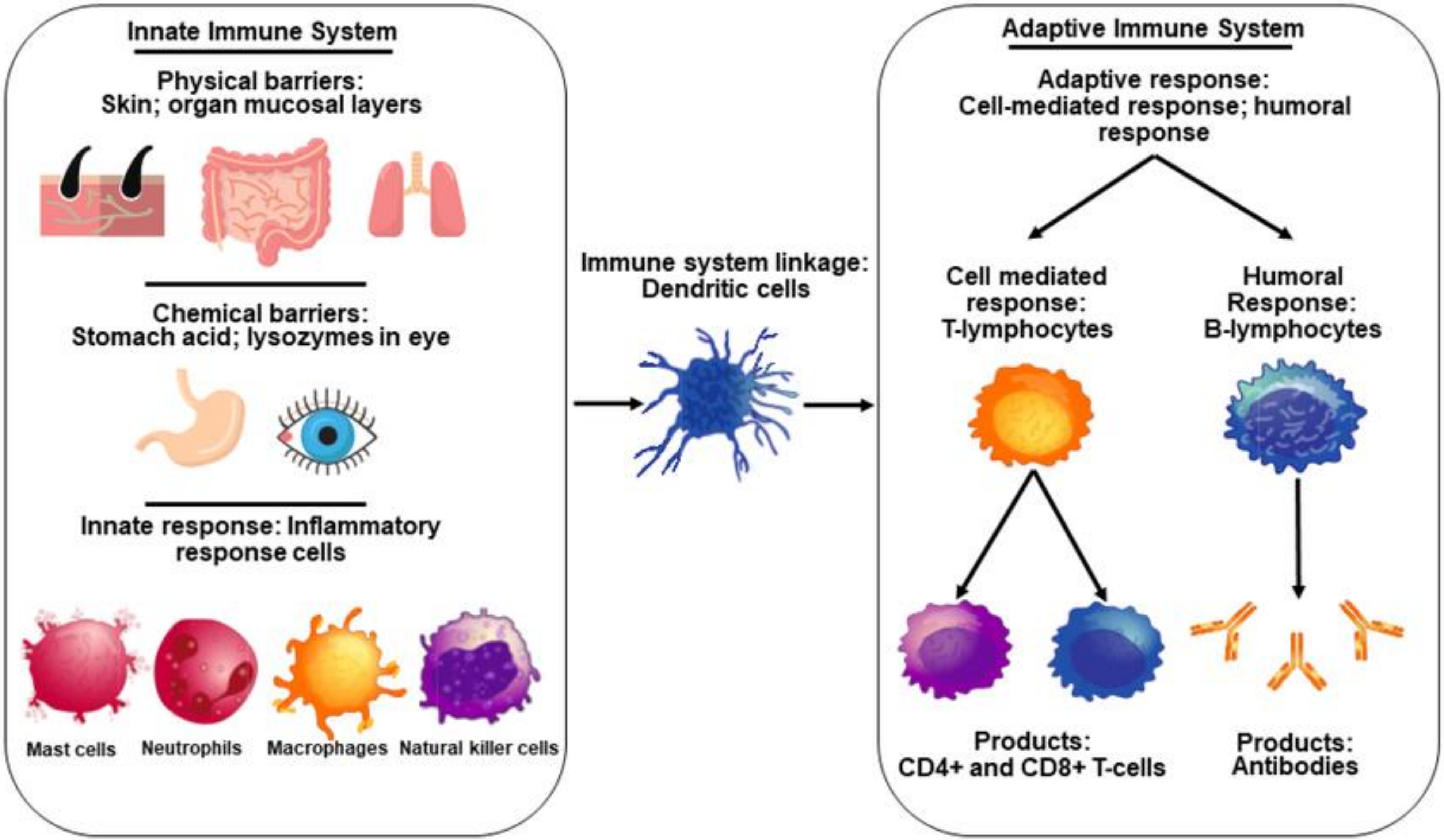
Type of immunity that develops when a person's **immune system responds to a foreign substance or microorganism, such as after an infection or vaccination.** Adaptive immunity involves specialized immune cells and antibodies that attack and destroy foreign invaders and are able to prevent disease in the future by remembering what those substances look like and mounting a new immune response. Adaptive immunity may last for a few weeks or months or for a long time, sometimes for a person's entire life.

It is important to note that we are born with our innate immune system almost fully developed. This is in contrast to our adaptive immune system which develops later and grows with us as we age and conquer new infections over time.

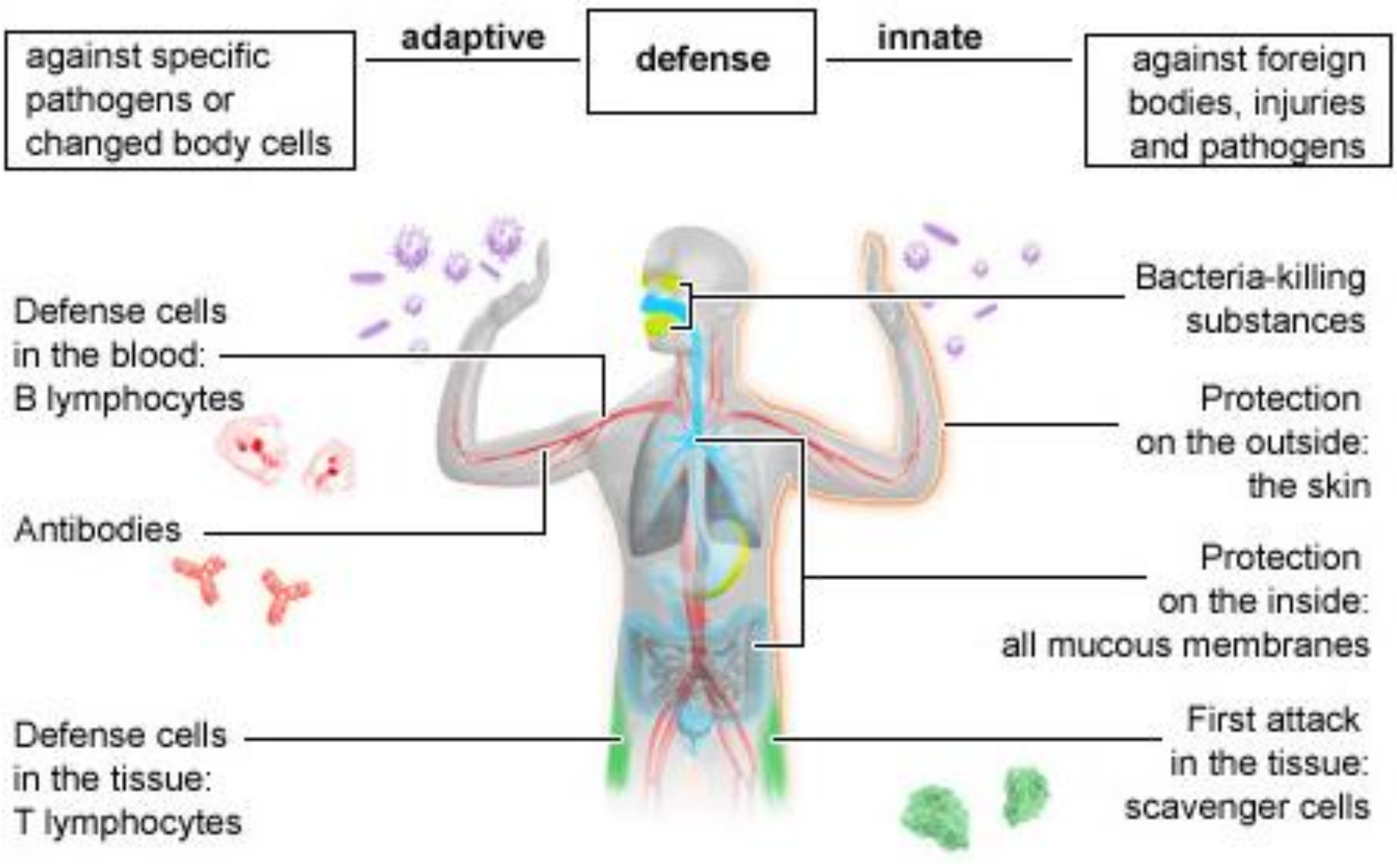
- adaptability, developed after the exposition to infection
- the immune response is not inherited, immunological response
- B and T cells, immunoglobulins, Antibodies.



Human immune system:



Innate Vs Adaptive immunity:



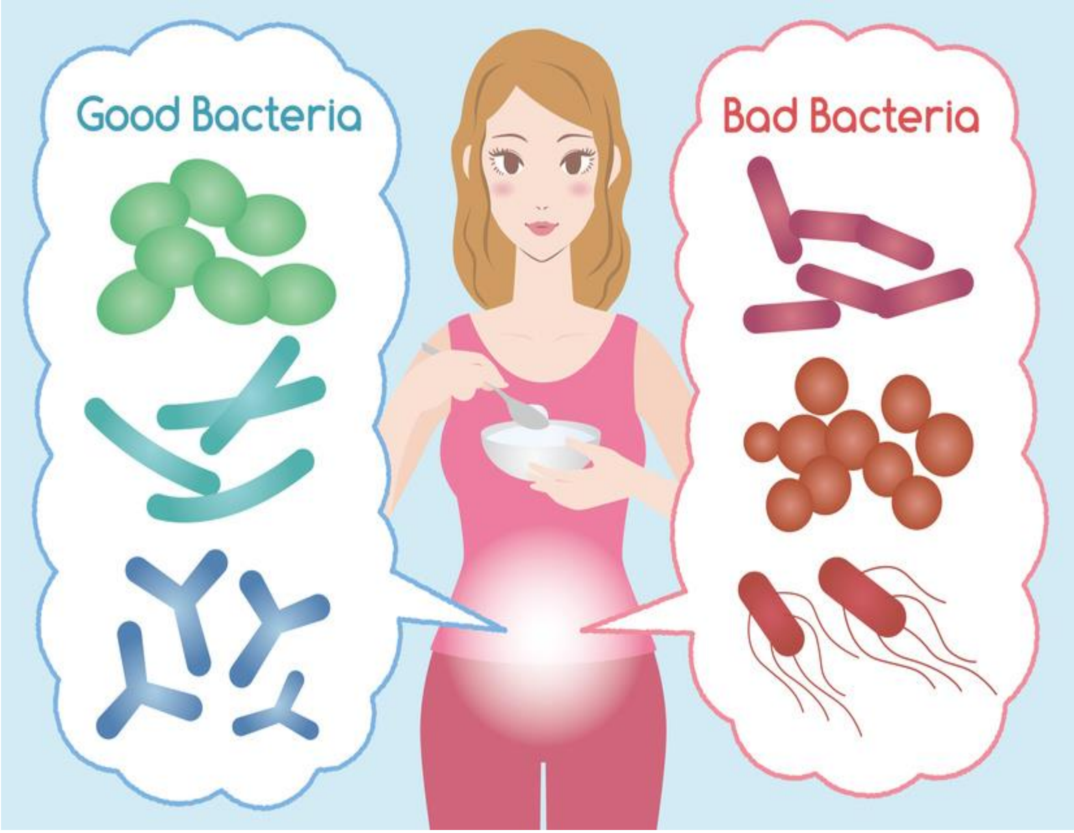
The immune system and microbial infection

- The immune system:

- Recording: keeps a record of every microbe it has ever defended, in **types of white blood cells (B- and T-lymphocytes)** known as **memory cells.**
- This means it can recognise and destroy the microbe quickly if it enters the body again, before it can multiply and make you feel sick.

Exception:

Some infections, like the flu and the common cold, many times as different strains- cause these illnesses

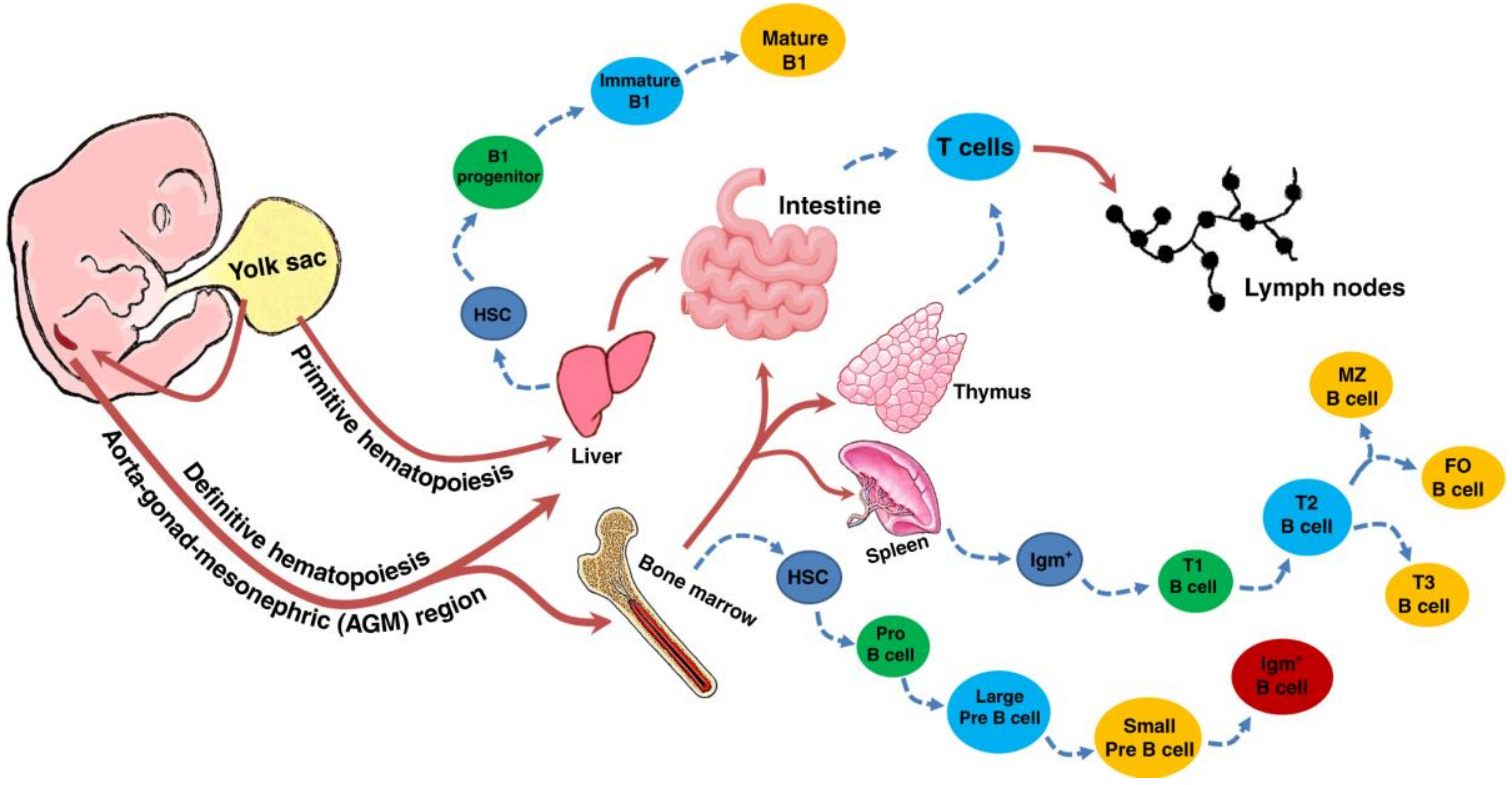


The body's other defences against microbes

As well as the immune system, the body has several other ways to defend itself against microbes, including:

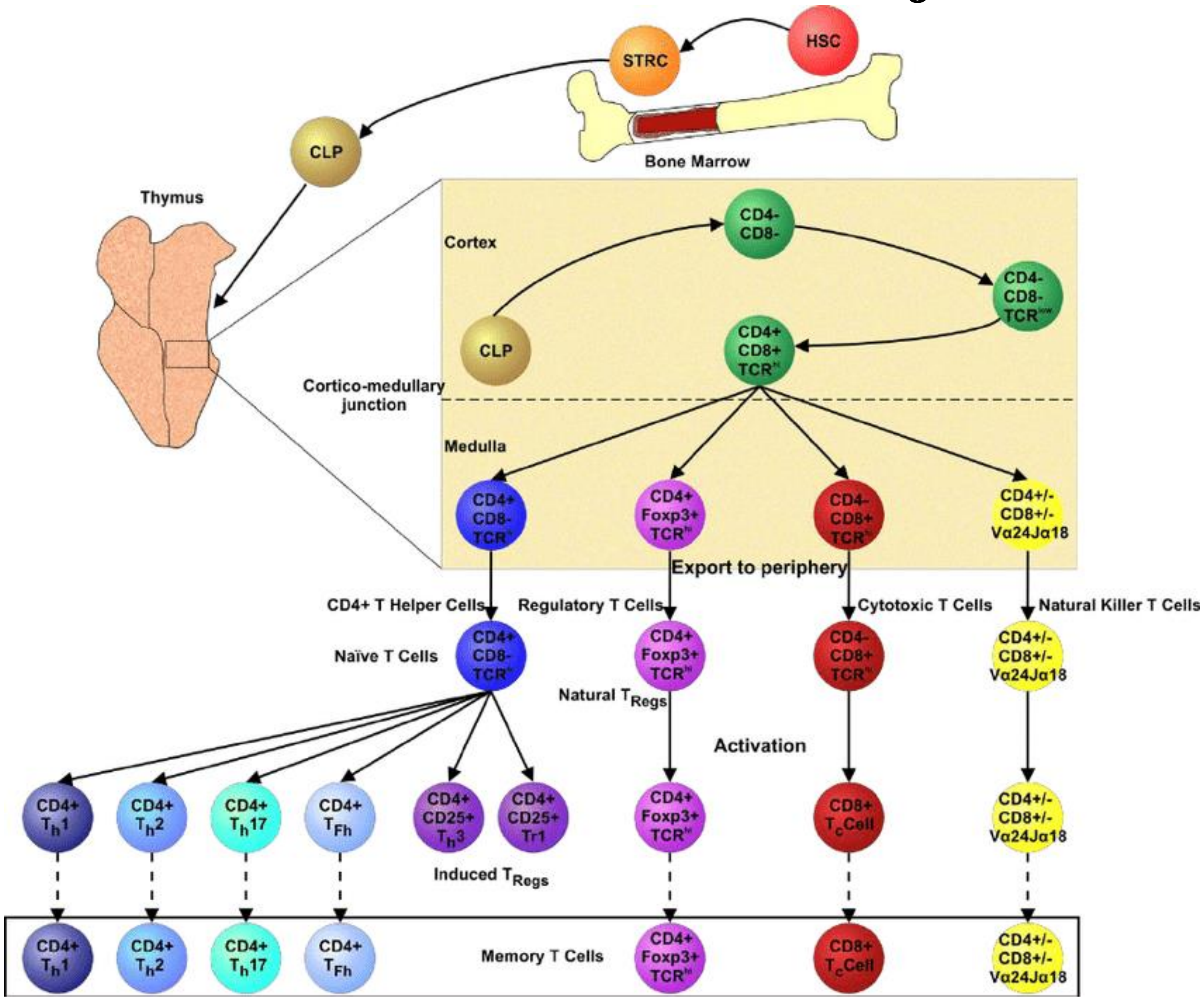
- **Skin: -** a waterproof barrier that secretes oil with bacteria-killing properties
- **Lungs:** - mucous in the lungs (phlegm) traps foreign particles, and small hairs (cilia) wave the mucous upwards so it can be coughed out
- **Digestive tract:** - the mucous lining contains antibodies, and the acid in the stomach can kill most microbes
- **Other defences:** - body fluids like **skin oil**, **saliva** and **tears** contain anti-bacterial enzymes that help reduce the risk of infection. The constant flushing of the **urinary tract** and the **bowel** also helps.

Development of the major immune cells in the human embryo.



Primitive hematopoiesis begins around 3rd week of gestation in the human yolk sac. Around 5th week of gestation, hematopoietic stem cells appear in the aorta-gonad-mesonephros (AGM) region and give rise to the hematopoietic progenitors that initiate definitive hematopoiesis in organs such as the liver and bone marrow. Brown marrows are indicating the migration route of hematopoietic cells and their progenitors (and precursors) among the hematopoietic organs throughout the fetal life. Broken blue arrows are indicating the synthesis of the immune cells from the vital organs.

Differentiation of major immune cells



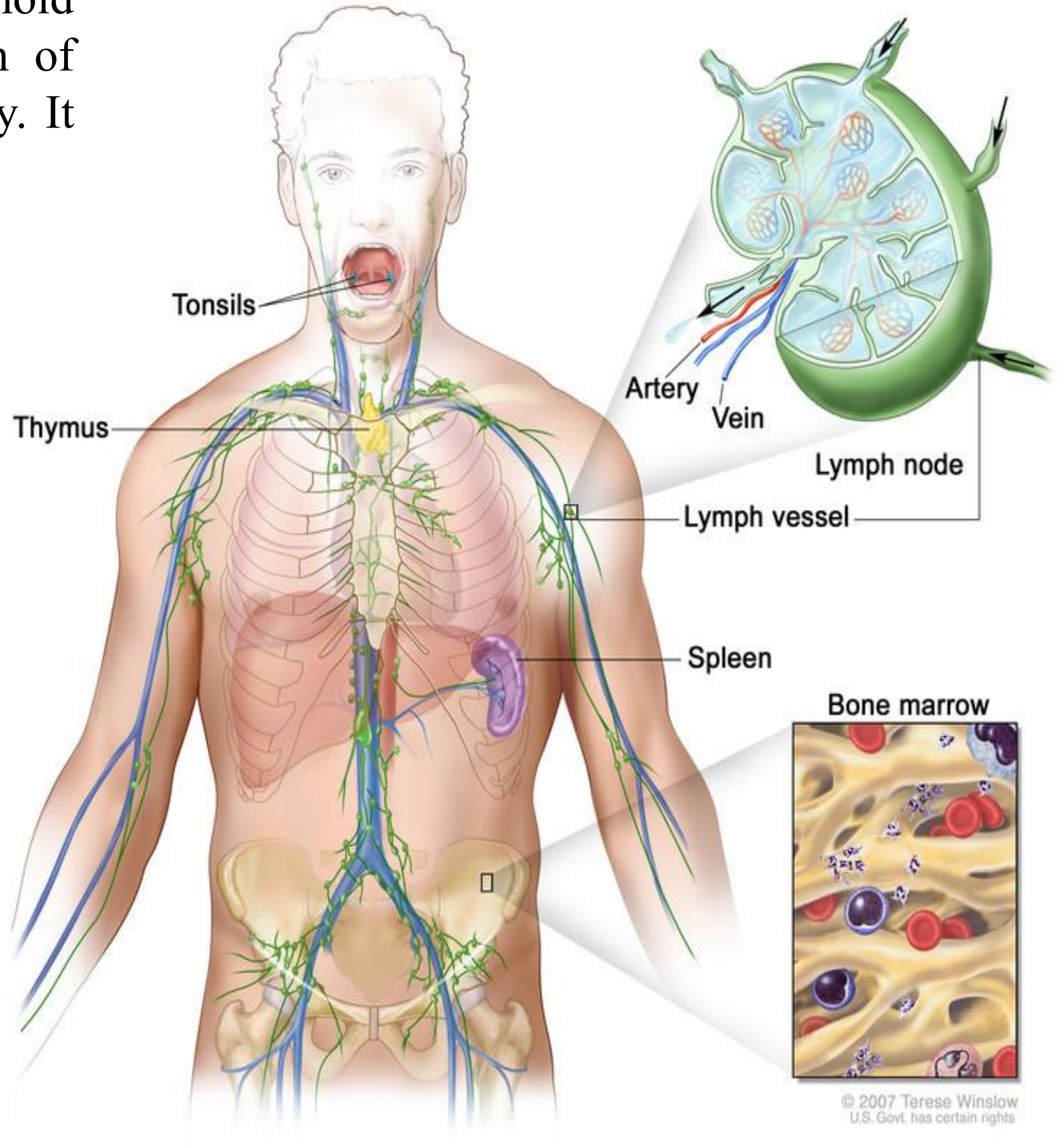
Differentiation of HSC derived precursor cells to multiple cell types within the thymus. Within the bone marrow of long bones, short term repopulating cells (STRC), derived from HSC differentiate to the common lymphoid progenitor (CLP) phenotype prior to entering the blood. CLPs subsequently migrate to the thymus, entering at the cortico-medullary junction, before undergoing a controlled series of molecular changes, the most significant being the expression of the T cell receptor (TCR), whilst passing through, and interacting with, the defined microarchitecture of the thymic cortex. Once the developing T cells express a TCR they migrate to the thymic medulla, at this stage they are known as double positive thymocytes because of their CD4 + CD8 + phenotype. In the medulla developing

Secondary lymphoid organs

The secondary lymphoid organs are also known as peripheral lymphoid organs. The function of secondary lymphoid organs is the activation of immune cells and to start a fight against foreign substances in our body. It includes spleen, lymph nodes, tonsils and various mucosal membranes.

Migration of mature general B and T cells to secondary lymphoid organs:

- Lymph nodes
- Spleen
- Tonsils
- External body surfaces (intestinal, respiratory, urinary, reproductive)

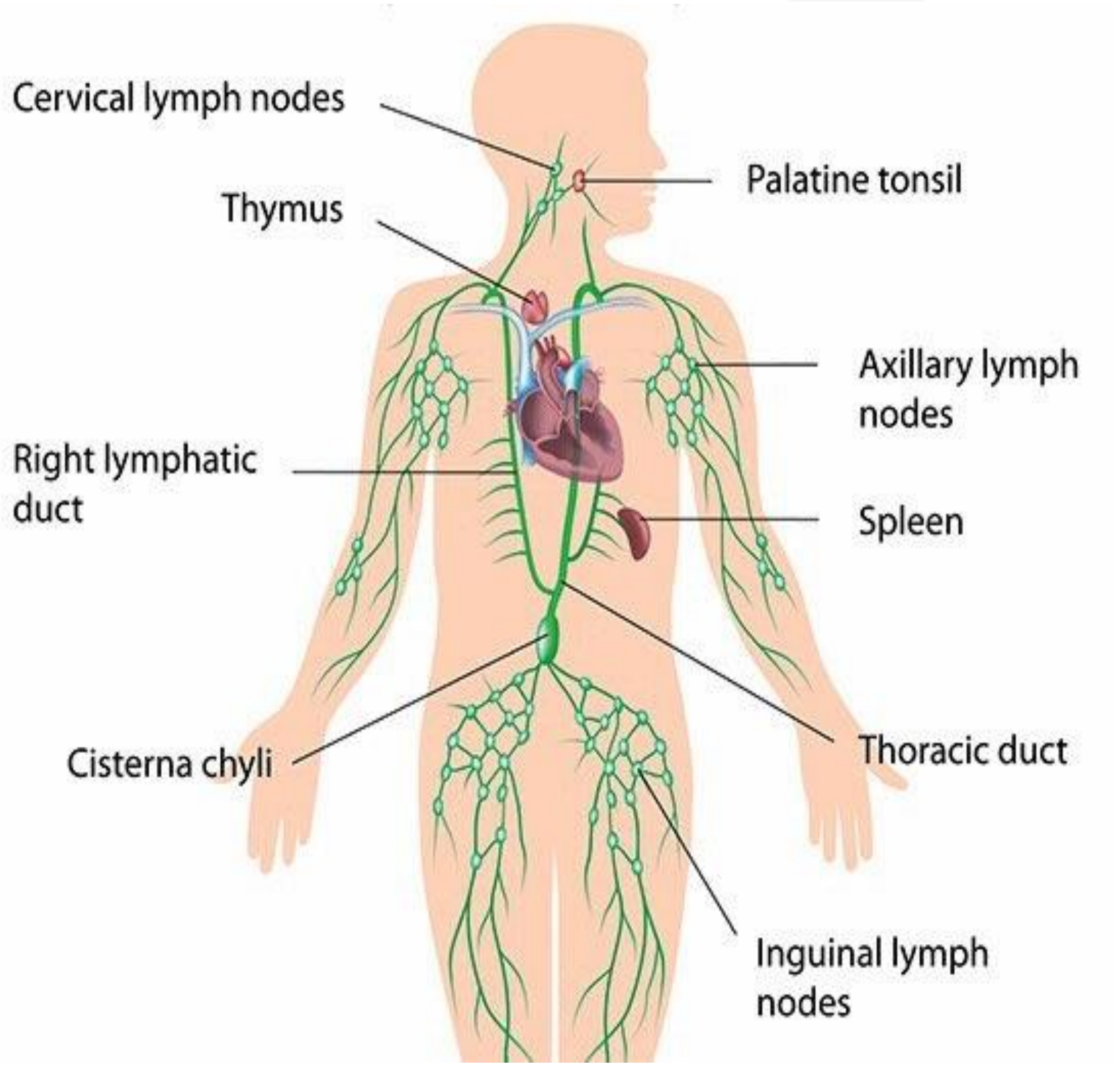


Lymphatic system:

- The lymphatic system is a network of delicate tubes throughout the body.
- manage the fluid levels in the body
- react to bacteria
- deal with cancer cells
- deal with cell products that otherwise would result in disease or disorders
- absorb some of the fats in our diet from the intestine.

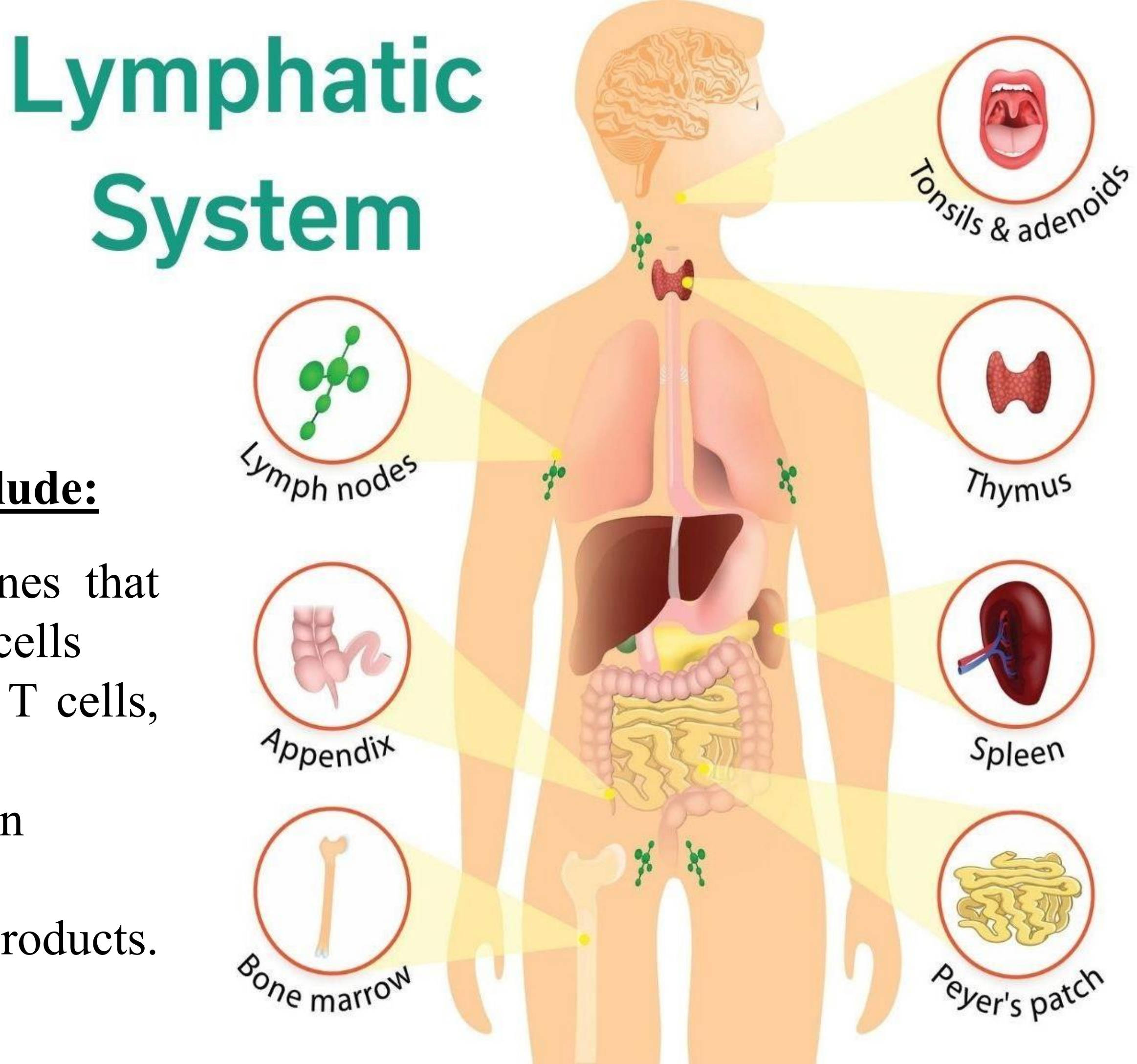
The lymphatic system is made up of:

- lymph nodes (also called lymph glands) -- which trap microbes
- lymph vessels -- tubes that carry lymph, the colourless that bathes your body's tissues and contains infection-fighting white blood cells
- white blood cells (lymphocytes).



The Immune Tissue: lymphatic system

The lymphatic system is part of the **immune system**. It keeps body fluid levels in balance and defends the body against infections. Lymphatic (lim-FAT-ik) vessels, tissues, organs, and glands work together to drain a watery fluid called **lymph** from throughout the body.



Organs and tissues that are part of the lymphatic system include:

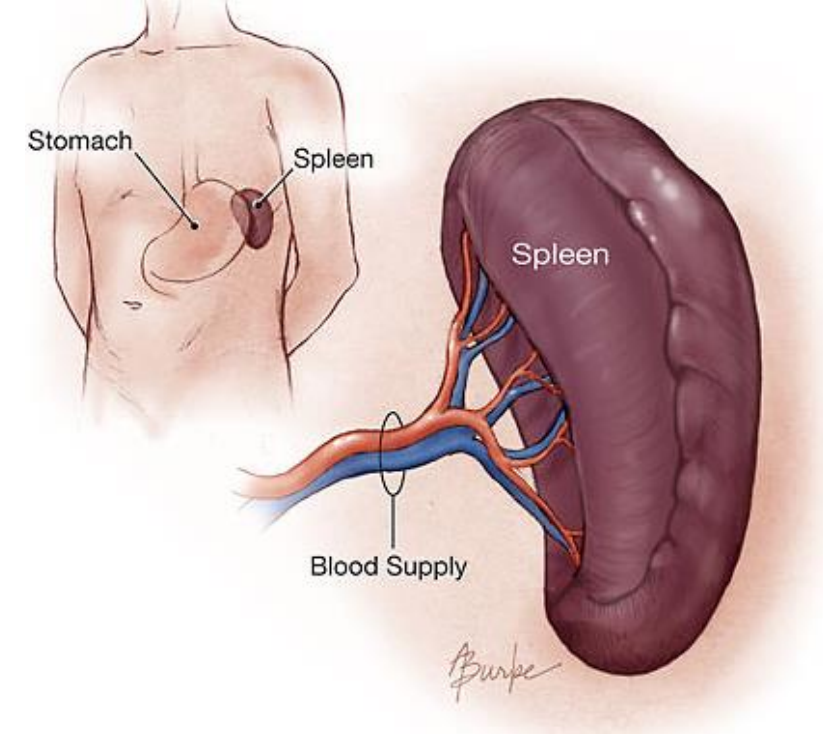
- **Bone marrow,** the thick, spongy kind of jelly inside bones that makes many kinds of blood cells, including immune system cells
- **Thymus gland,** which makes immune system cells called T cells, especially before and during puberty
- **Tonsils** filter out bacteria and other germs to prevent infection
- **Appendix**-store certain healthy types of gut bacteria
- **Lymph nodes:** filter out harmful substances and waste products. reserves immune cells (lymphocytes)

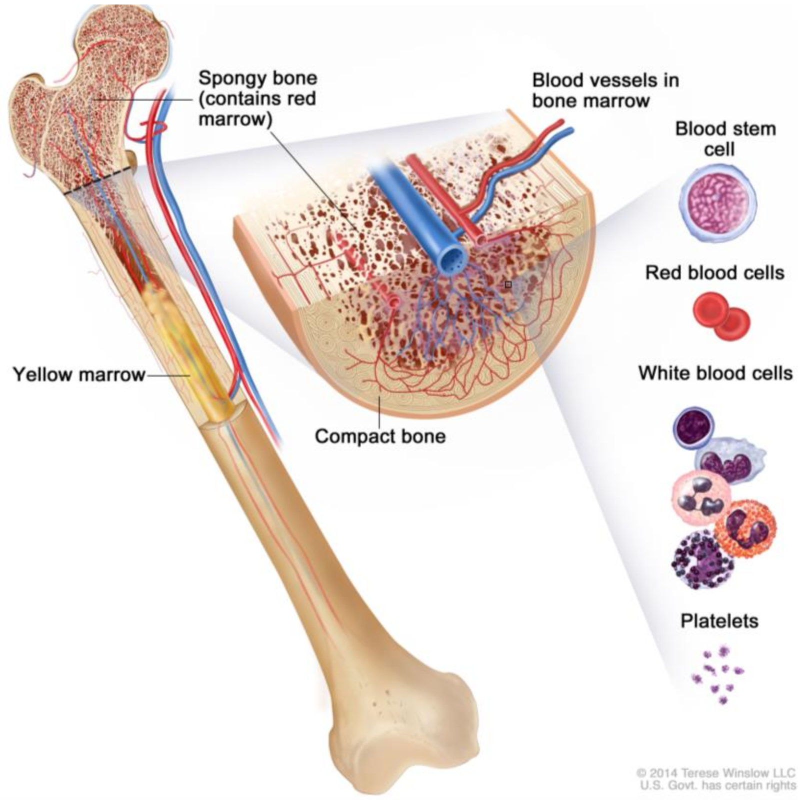
https://kidshealth.org/en/parents/spleen-lymphatic.html

The Immune Tissue: Spleen and Bone Marrow

The spleen is part of the **lymphatic system**, which is an extensive drainage network. The lymphatic system keeps body fluid levels in balance and to defend against infections. It is made up of a network of lymphatic vessels that carry **lymph** a clear, watery fluid that contains proteins, salts, and other substances throughout the body. Mainly function is:

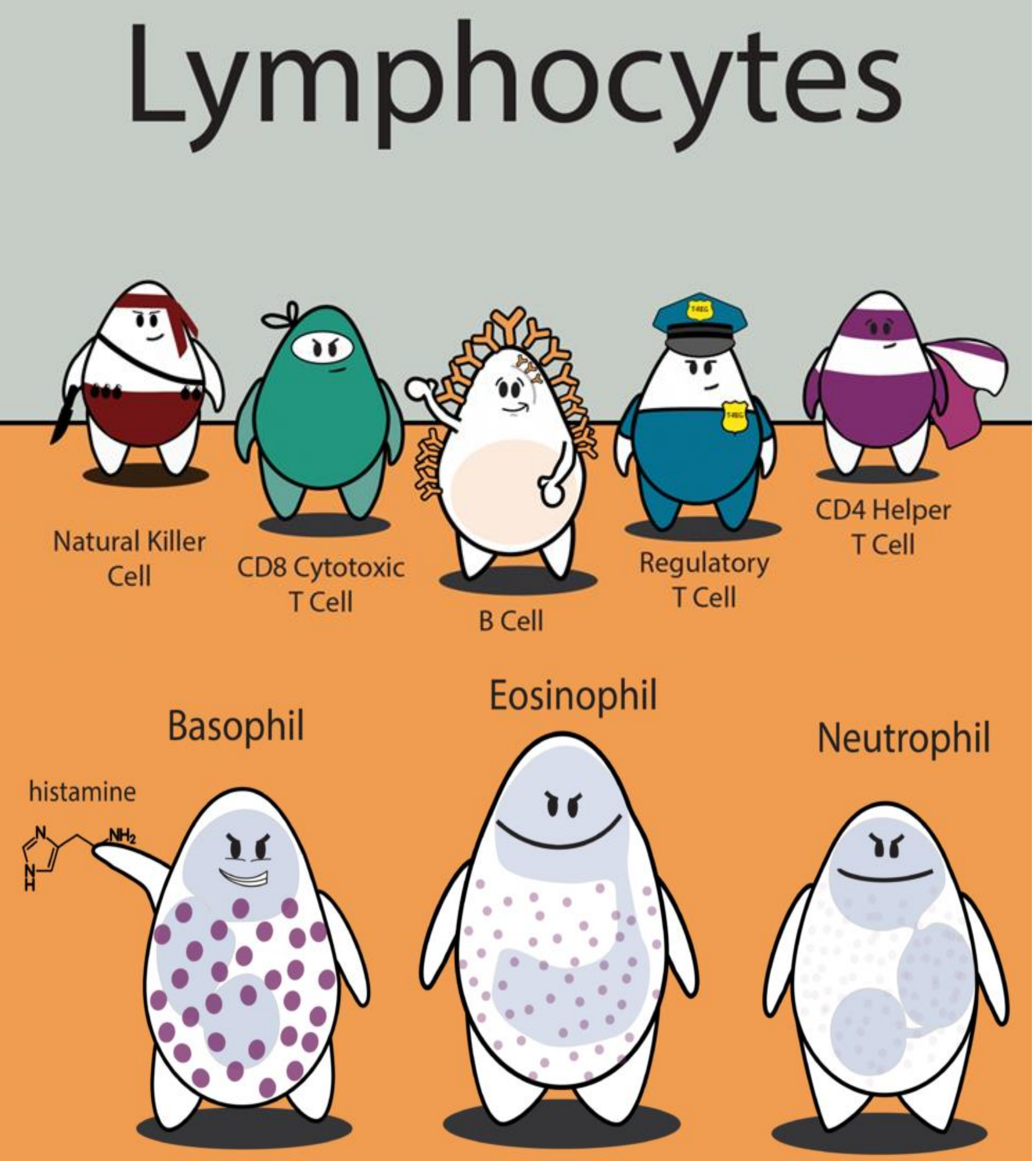
•Stores blood.
•Filters blood by removing cellular waste and getting rid of old or damaged blood cells.
•Makes white blood cells and antibodies that help you fight infection.
•Maintains the levels of fluid in your body.
•Produces antibodies that protect you against infection.





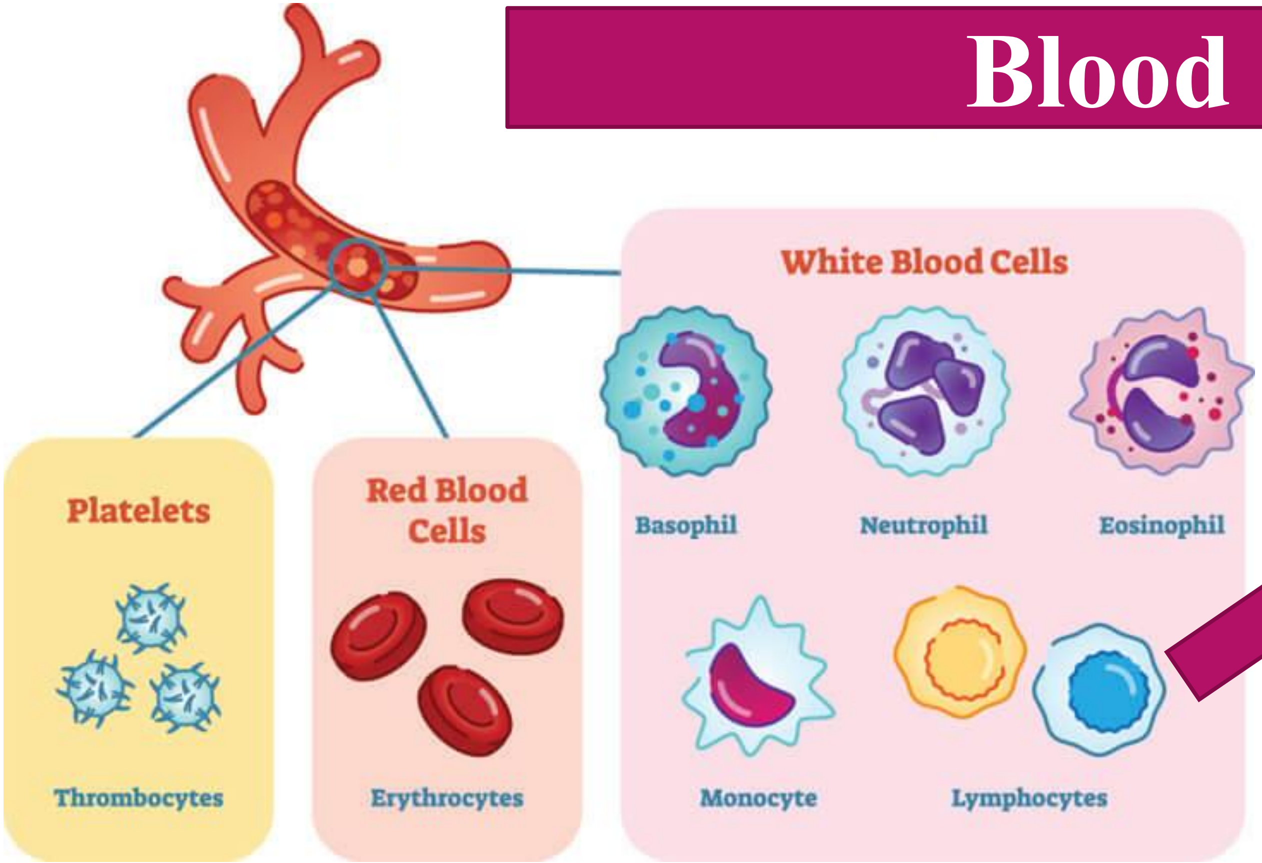
The soft, spongy tissue that has many blood vessels and is found in the center of most bones. There are two types of bone marrow: red and yellow. Red bone marrow contains blood stem cells that can become red blood cells, white blood cells, or platelets. Yellow bone marrow is made mostly of fat and contains stem cells that can become cartilage, fat, or bone cells.

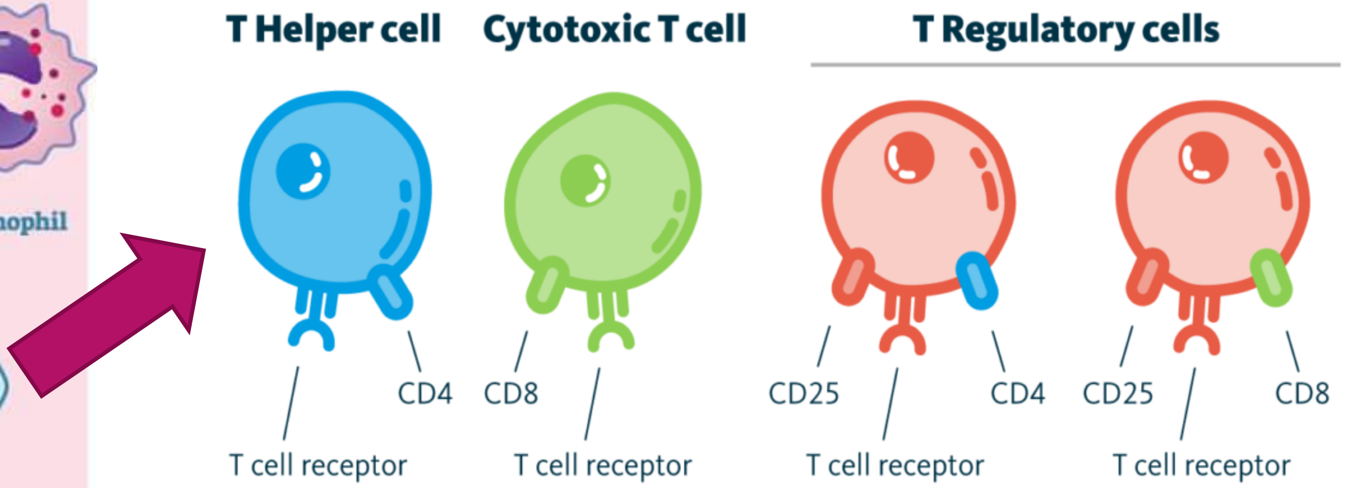
- It is the "Blood cell factory". Produces and releases blood cells into the bloodstream when they are mature and when required.
- Produces the red blood cells to carry oxygen,
- Produces WBC to fight infection, and
- the platelets blood clot.



Immune Cells:

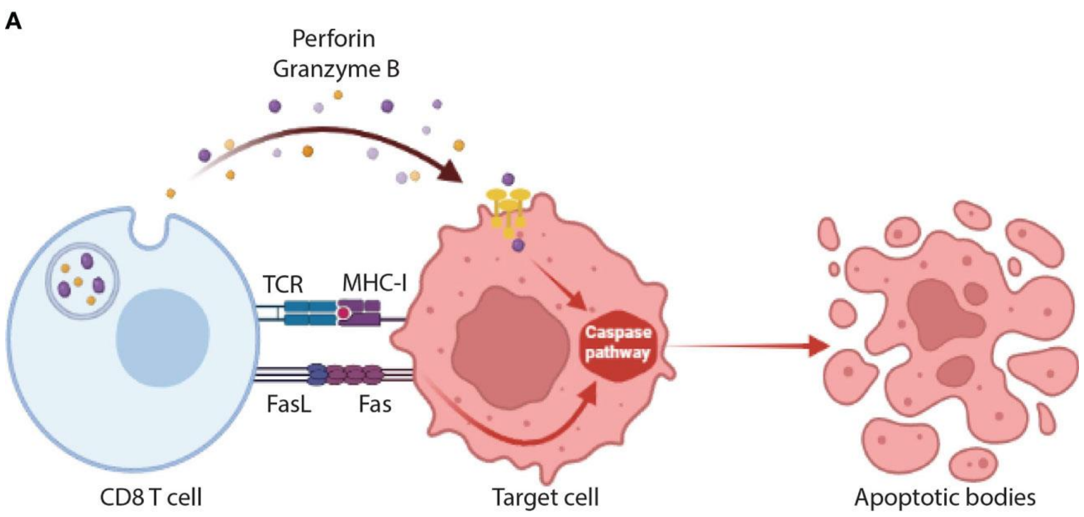
Blood cells

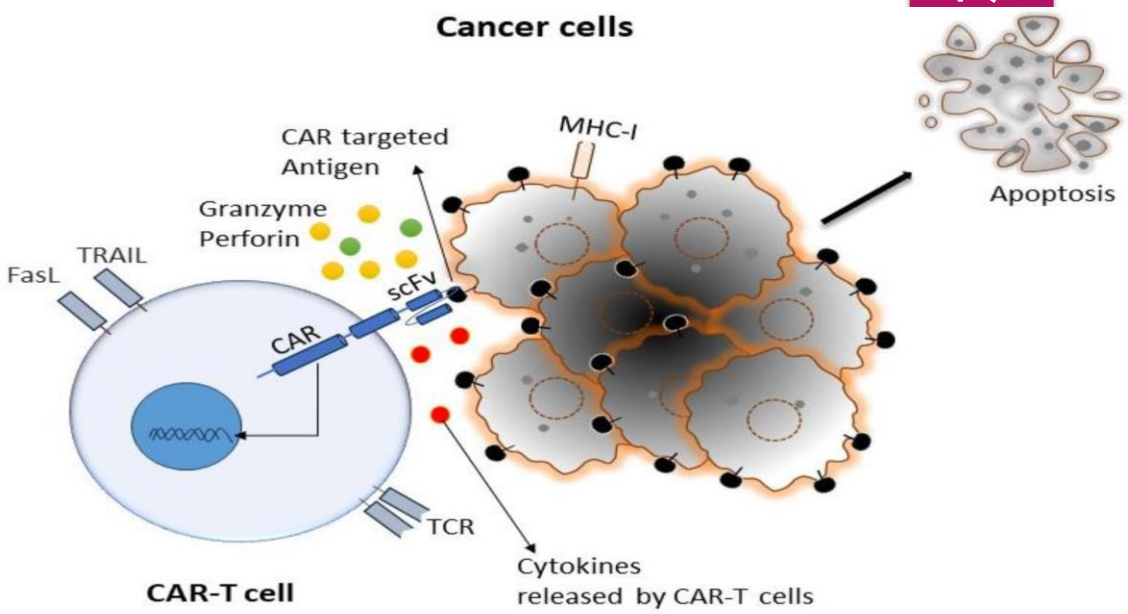




- White blood cells are the key players in your immune system.
- Formation: bone marrow and other part of the lymphatic system.
- Process: White blood cells move through blood and tissue throughout our body, looking for foreign invaders (microbes) such as bacteria, viruses, parasites and fungi. When they find them, they launch an immune attack.
- White blood cells include lymphocytes (such as B-cells, T-cells and natural killer cells), and many other types of immune cells.

Mechanism of CD8 T Cells mediated Killing



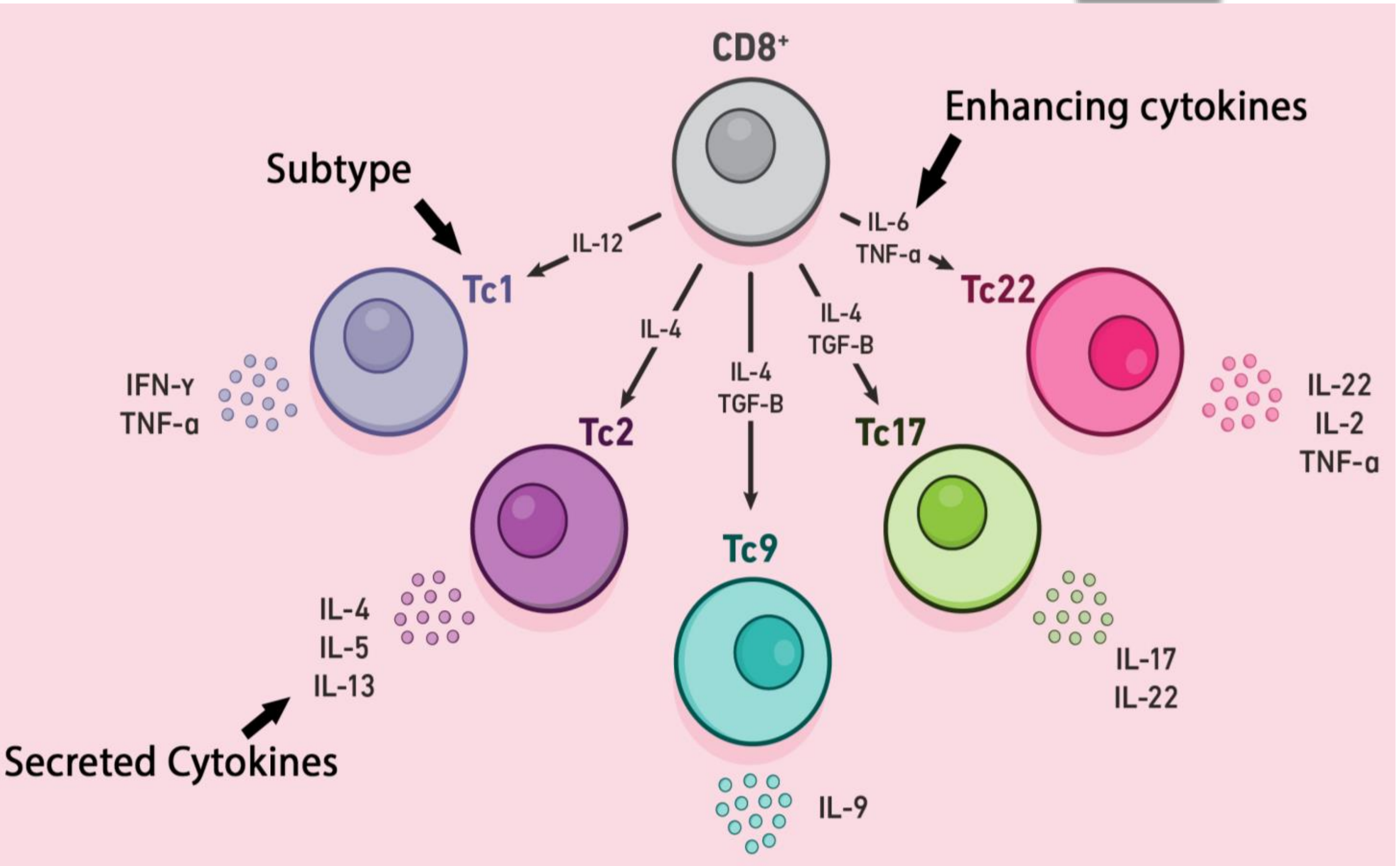


Mechanism of Action of Cytotoxic T Cells (CD8 cells/CTLs):

1. **Recognition:** CTLs use T-cell receptors (TCRs) to identify infected or abnormal cells by binding to antigenic peptides presented on MHC-I molecules.
2. **Activation:** Upon recognition, CTLs become activated and release cytotoxic granules.
3. **Target Cell Killing:**
    1. **Perforin** forms pores in the target cell membrane.
    2. **Granzymes** enter through these pores and induce apoptosis.
    3. **Fas-FasL Pathway:** FasL on CTLs binds to Fas on target cells, triggering apoptosis.

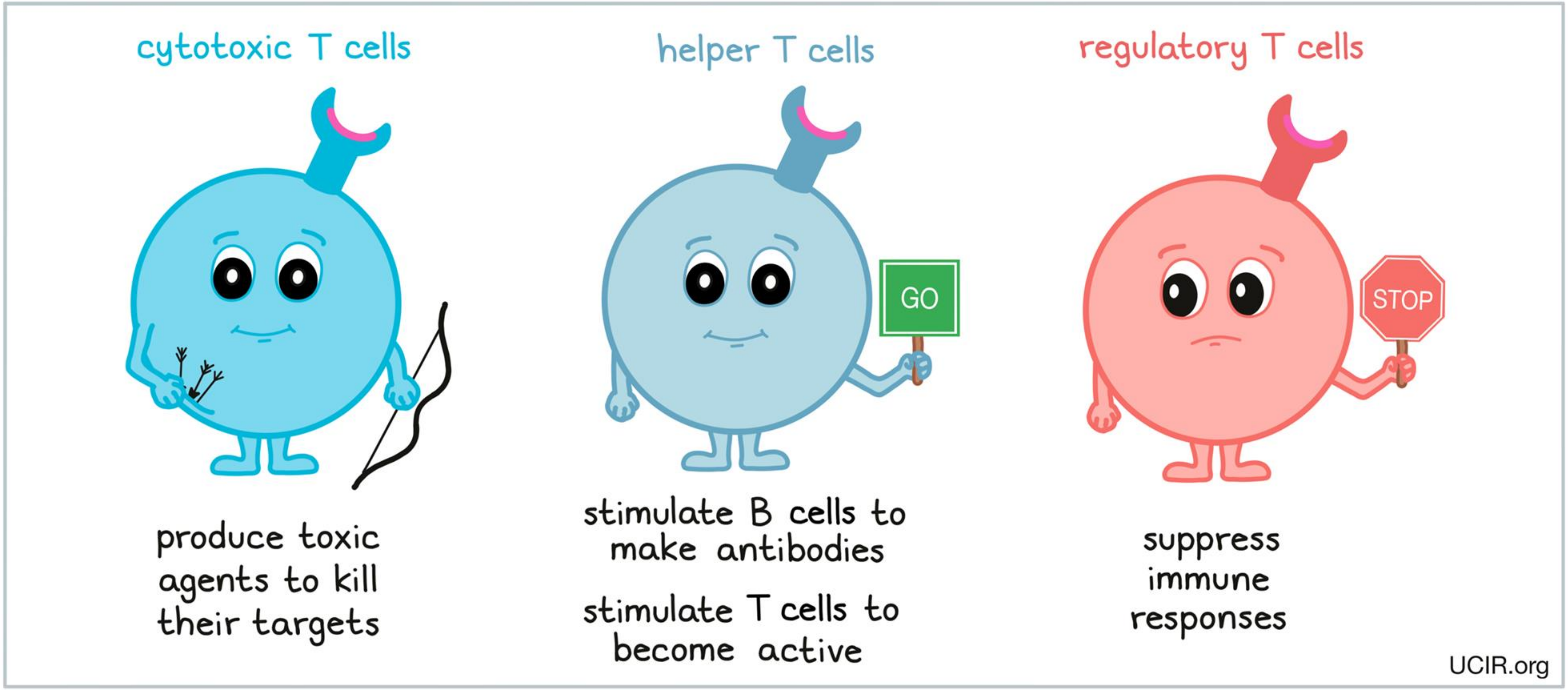
Role of Cytotoxic CD8 T Cells in CD4+ T Cell Activation:

1.Cytokine Release:
Produce **IL-2** and other cytokines that promote CD4+ and CD8+ T cell growth and differentiation.
2. Enhancing Antigen Presentation:
Indirectly aid APC "licensing" by CD4+ T cells, boosting APC ability to activate CD8+ T cells and reinforcing CD4+ activation.
3. Direct Interaction: May directly influence CD4+ T cells via surface molecules or cytokine signaling.



Summary:
Cytotoxic T cells support CD4+ T cell activation through cytokine production, APC modulation, enhanced antigen presentation, and direct cellular interactions.

Main Function of White blood cells



Natural Killer (NK) cells activate other Immune cells

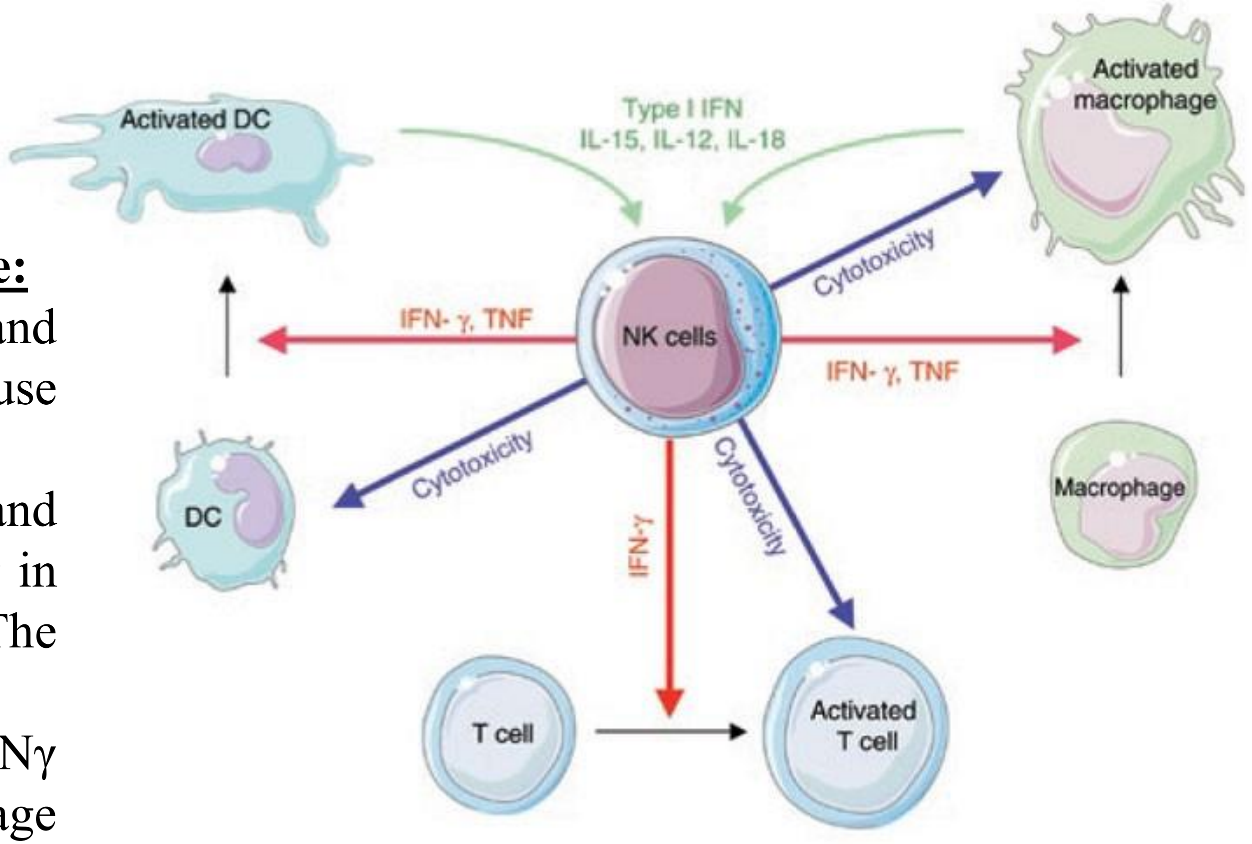
Natural killer cells (NK cells) are white blood cells that destroy infected cells and cancer cells in our body. NK cells are important fighters in our immune system. Our immune system protects us from harmful invaders, like pathogens (viruses, bacteria and parasites) and cancer cells.

Natural killer cells activate to destroy target cells include:

•**Cells that release activating signals**: Cancer cells and infected cells sometimes release chemical signals that cause NK cells to attack.

•**Activate other Immune cells**: NK cells release perforin and granzymes to kill a target cell. Perforin creates an opening in the target cell so the NK cell can insert granzymes. The granzyme kills the cell.

•Activated NK cells NK cells secrete cytokines such as IFNγ and TNFα, which act on other immune cells like Macrophage and Dendritic cells to enhance the immune response.



Mast Cells

- One type of white blood cell that is found in connective tissues all through the body, especially under the skin, near blood vessels and lymph vessels, in nerves, and in the lungs and intestines.
- They contain chemicals such as histamine, heparin, cytokines, and growth factors. They release these chemicals during allergic reactions and certain immune responses.
- These chemicals have many effects, including the widening of blood vessels and angiogenesis. During an allergic response, they can cause flushing (a hot, red face) and itching.

This occurs when damaged tissue and white blood cells (basophils), release histamine. Histamines, released from mast cells, cause the blood vessels to dilate allowing the tissue to become more permeable to tissue fluid.

- Inflammation is characterized by four symptoms:
  - Redness
  - Swelling
  - Pain
  - Heat

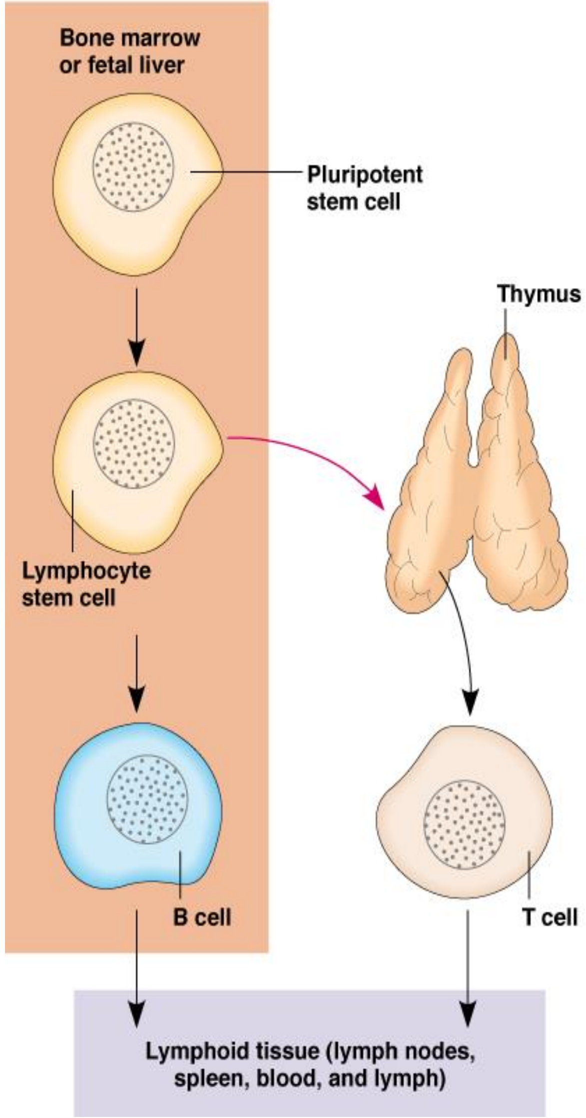
The immunological mechanisms of inflammation:

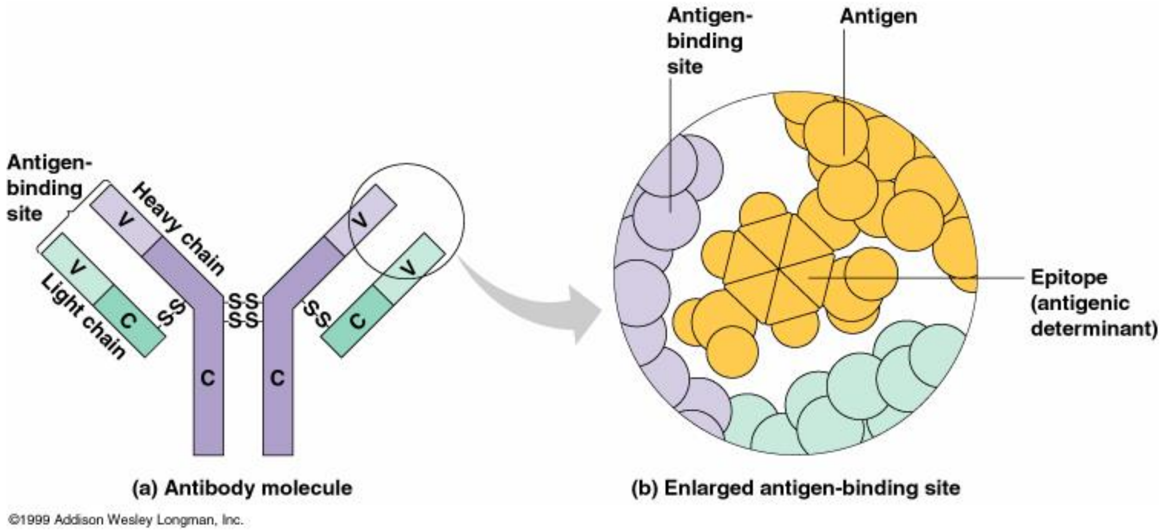
Acute inflammation is characterized by vasodilatation, fluid exudation and neutrophil infiltration. These processes are activated and amplified by a series of intracellular and extracellular factors that tightly co-ordinate the inflammatory process. The innate immune system responds rapidly to infection or injury.

- Cell mediated response – neutrophils, macrophages, lymphocytes accumulate together
- Fever and inflammatory cytokines production (IL-1, IL-10, TNF)
- Production of acute phase proteins
- Complement system activation
- Specific response (production of antigen-specific antibodies and T cells)

Antibodies are Produced by B Lymphocytes

Antibodies are Proteins that Recognize Specific Antigens





How Do B Cells Produce Antibodies?

- B cells develop from stem cells in the bone marrow of adults (liver of fetuses).
- After maturation B cells migrate to lymphoid organs (lymph node or spleen).
- Clonal Selection: When a B cell encounters an antigen it recognizes, it is stimulated and divides into many clones called plasma cells, which actively secrete antibodies.
- Each B cell produces antibodies that will recognize only one antigenic determinant.

Antibodies

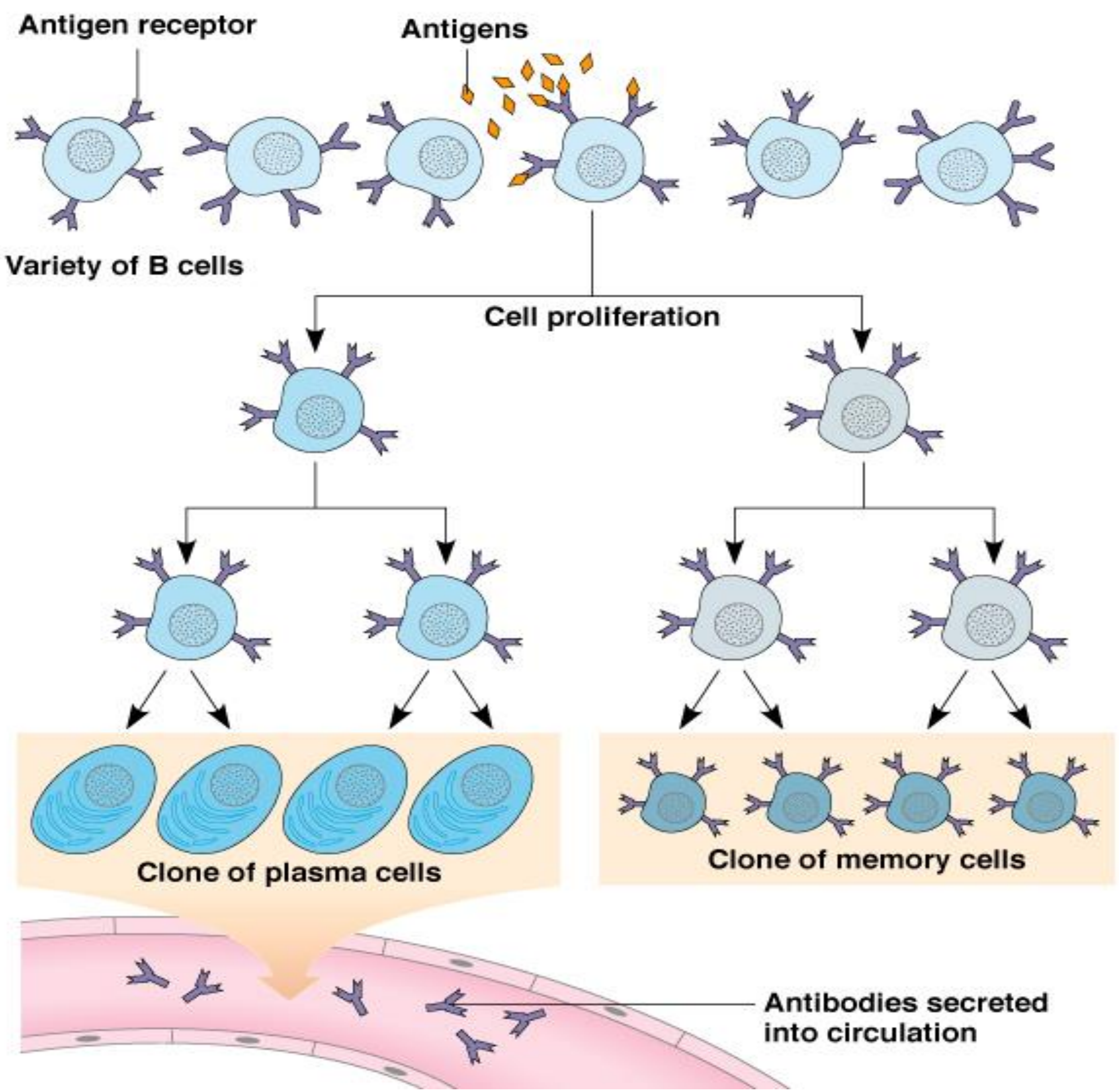
- Proteins that recognize and bind to a particular antigen with very high *specificity*.
- Made in response to exposure to the antigen.
- One virus or microbe may have several *antigenic determinant sites*, to which different antibodies may bind.
- Each antibody has at least two identical sites that bind antigen: *Antigen binding sites*.
- Belong to a group of serum proteins called immunoglobulins (Igs).

monoclonal vs. polyclonal antibody

Monoclonal Antibody: homogeneous antibody preparations produced in the laboratory. Consist of a single type of antigen binding site, produced by a single B cell clone.

Polyclonal antibodies: antibody preparations from immunized animals. Consist of complex mixtures of different antibodies produced by many different B cell clones

How B Cells produced antibody by Antigenic Stimulation

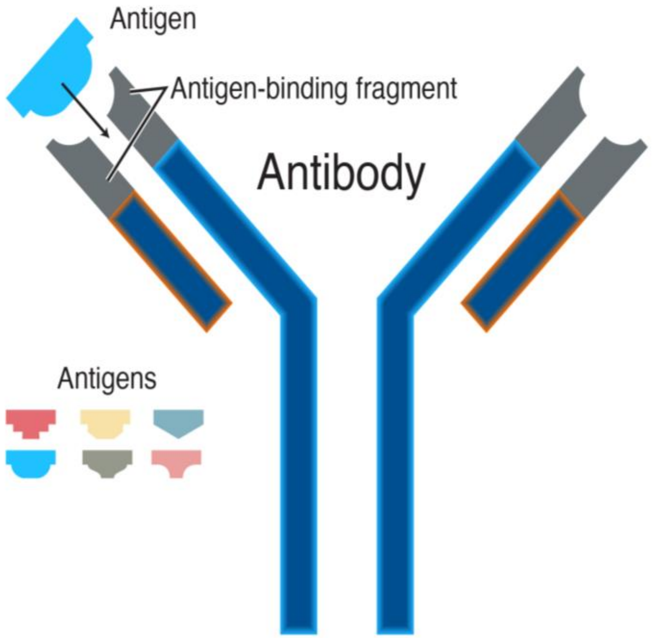


Once B cells are activated, they become plasma cells that produce antibodies in response to an antigen.

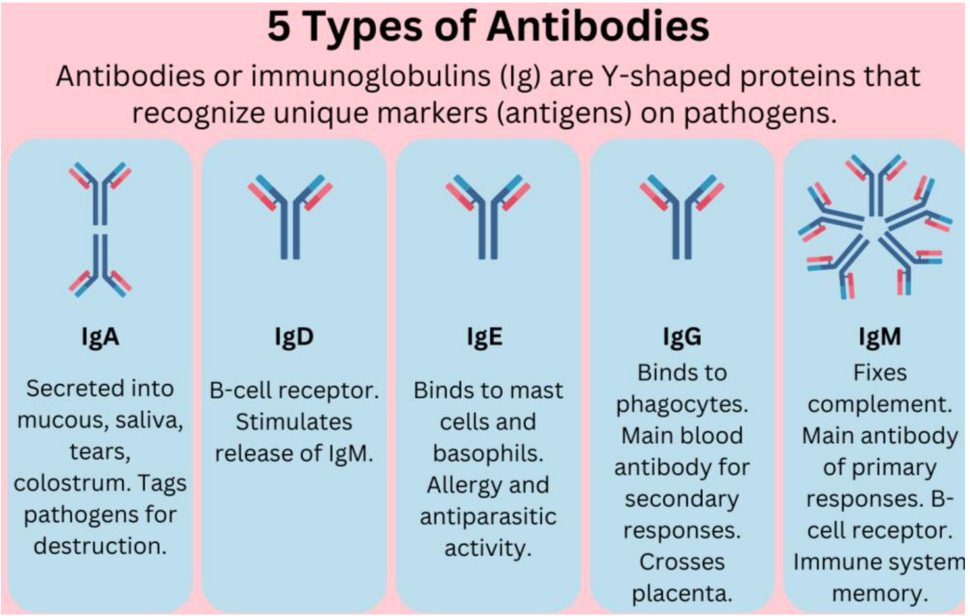
Or they become memory cells that remember the antigen so your immune system can quickly identify and fight it in the future.

Antibodies (Ab) / (immunoglobulin)

A protein made by plasma cells (a type of white blood cell) in response to an antigen (a substance that causes the body to make a specific immune response). Each antibody can bind to only one specific antigen.
The purpose of this binding is to help destroy the antigen. Some antibodies destroy antigens directly. Others make it easier for white blood cells to destroy the antigen. An antibody is a type of immunoglobulin.



- Recognize and bind with microbes/toxins/foreign pathogens/antigen/poisons/dead cells/cancer cells
- Treats as foreign: antigens on the surface of the microbes, or in the chemicals they produce, or toxin they bind and destroy destruction of foreign pathogens and neutralize them.
- There are 5 different types of antibodies, each with a different function, they are classified into IgG, IgM, IgA, IgD, and IgE
- Among them IgG is the most abundant antibody isotype in the blood (plasma), accounting for **70-75% of total** human antibodies.



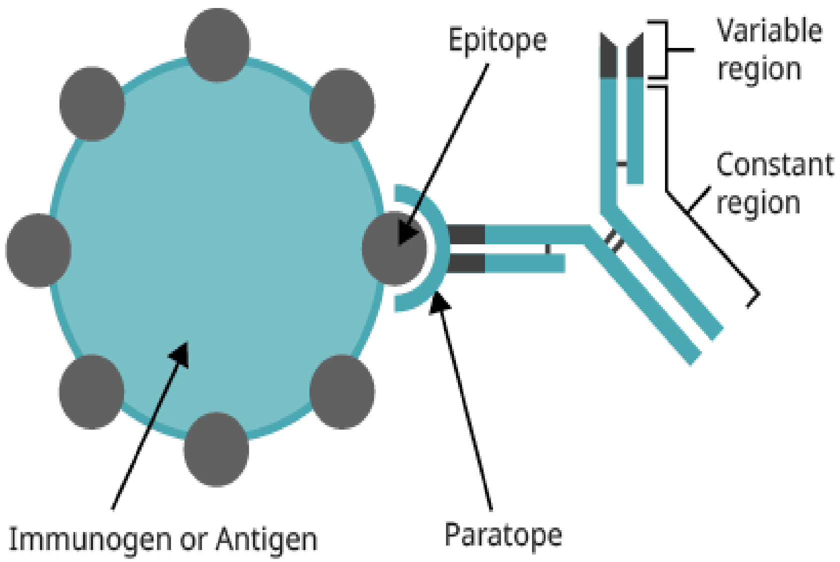
Antigens (Ag)

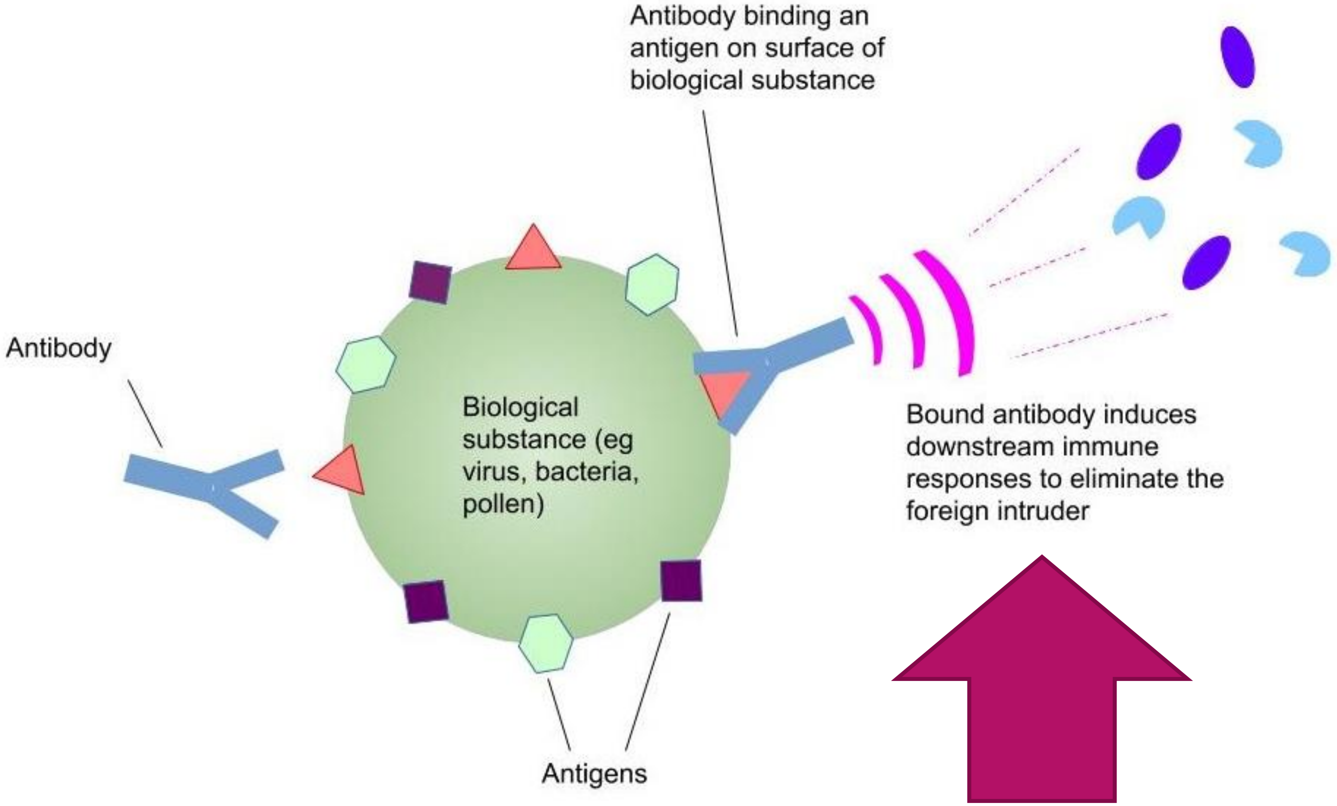
- Mostly **proteins or large polysaccharides from a foreign organism**.
  - **Cancer Antigens:** mutated peptides produced by cancer cells
  - **Microbes:** Capsules, cell walls, toxins, viral capsids, flagella, etc.
  - **Nonmicrobial:** Pollen , red blood cell surface molecules, serum proteins, and surface molecules from transplanted tissue.
- Lipids and nucleic acids are only antigenic when combined with proteins or polysaccharides.
- Molecular weight of 10,000 or higher.

Antigens (Ag) / Immunogen

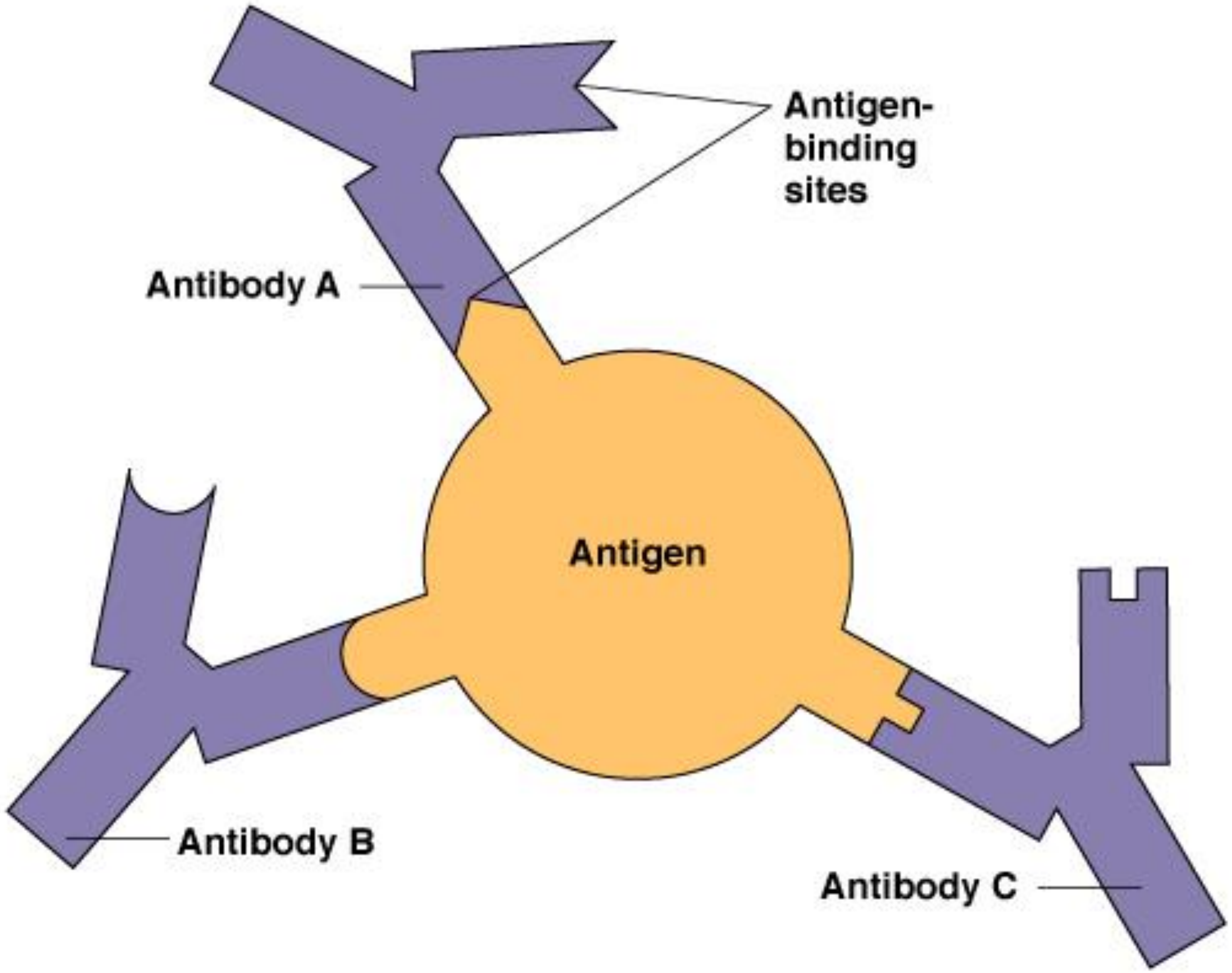
Epitope:

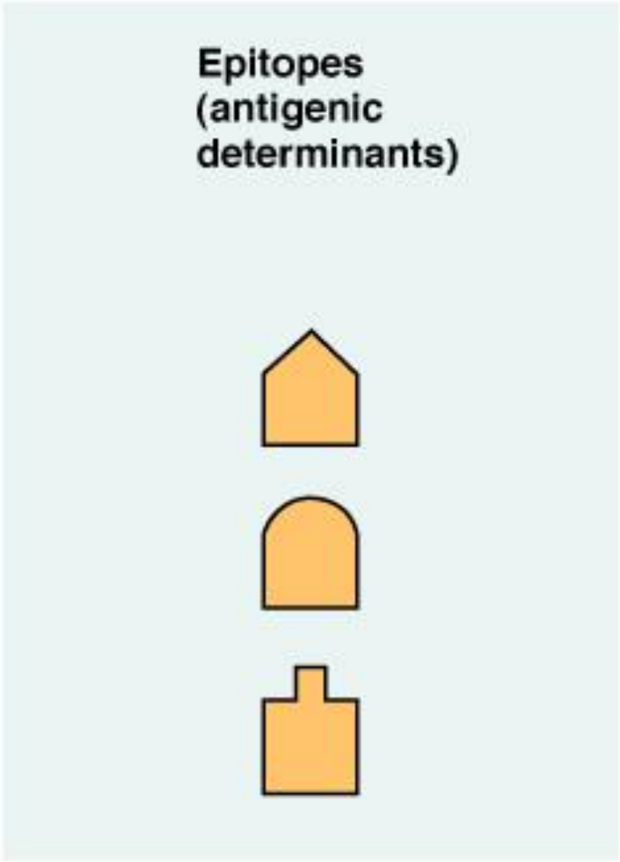
- Small part of an antigen that interacts with an antibody.
- Any given antigen may have several epitopes.
- Each epitope is recognized by a different antibody.





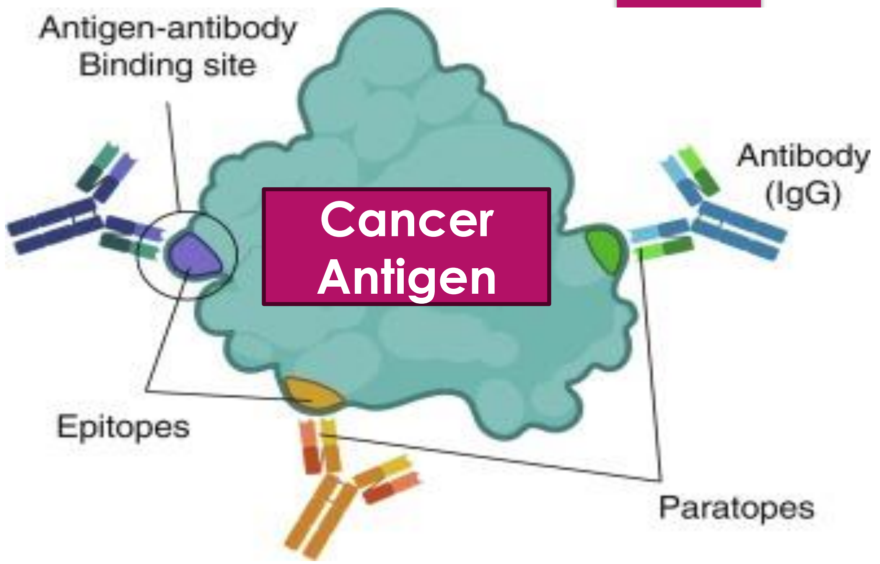
Epitopes: Antigen Regions that Interact with Antibodies



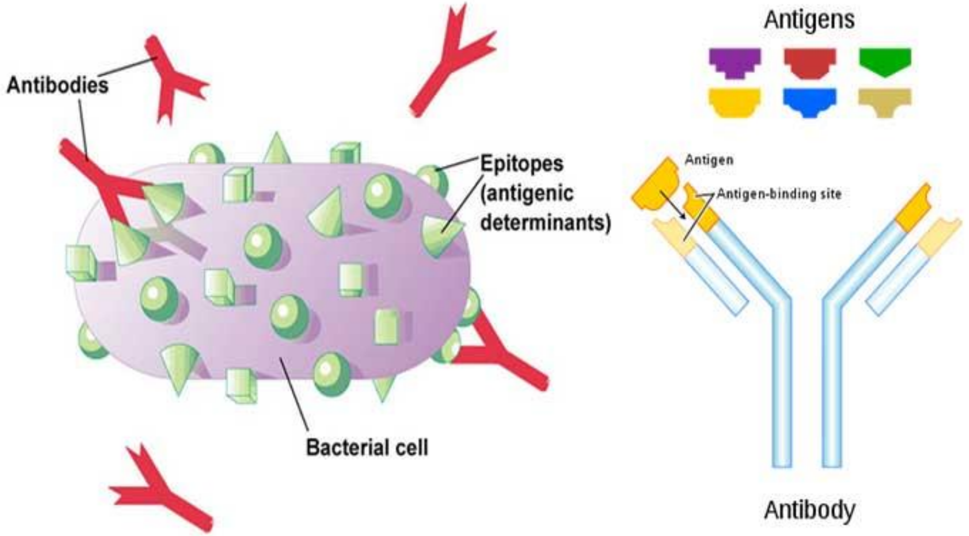


Different types of Antigens (Ag)

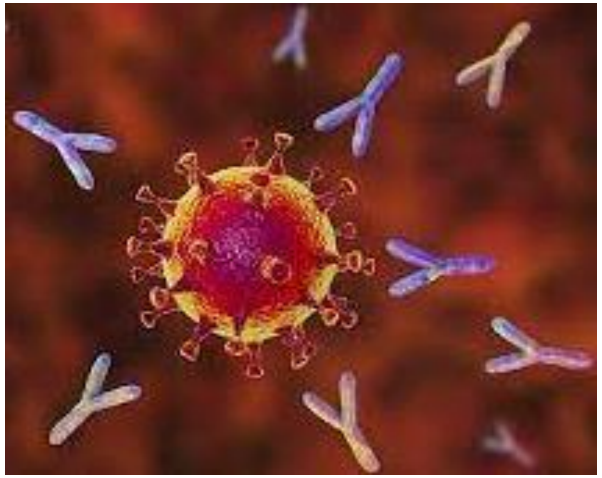
- Any substance that causes the body to make an immune response against that substance. Antigens include toxins, chemicals, bacteria, viruses, or other substances that come from outside the body.
- Body tissues and cells, including cancer cells, also have antigens on them that can cause an immune response.
- These antigens can also be used as markers in laboratory tests to identify those tissues or cells.



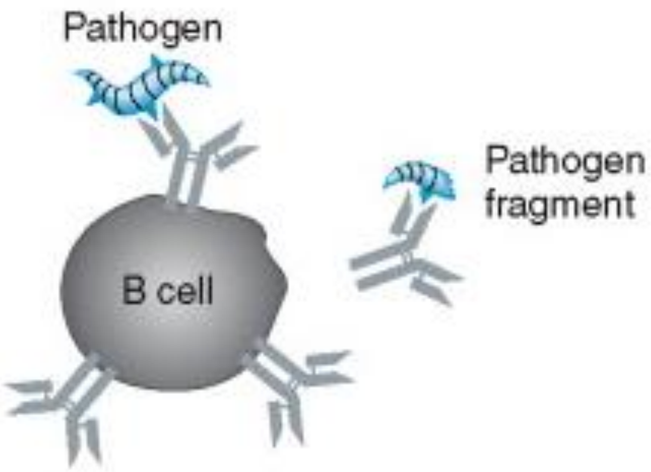
Bacterial Ag



Viral Ag

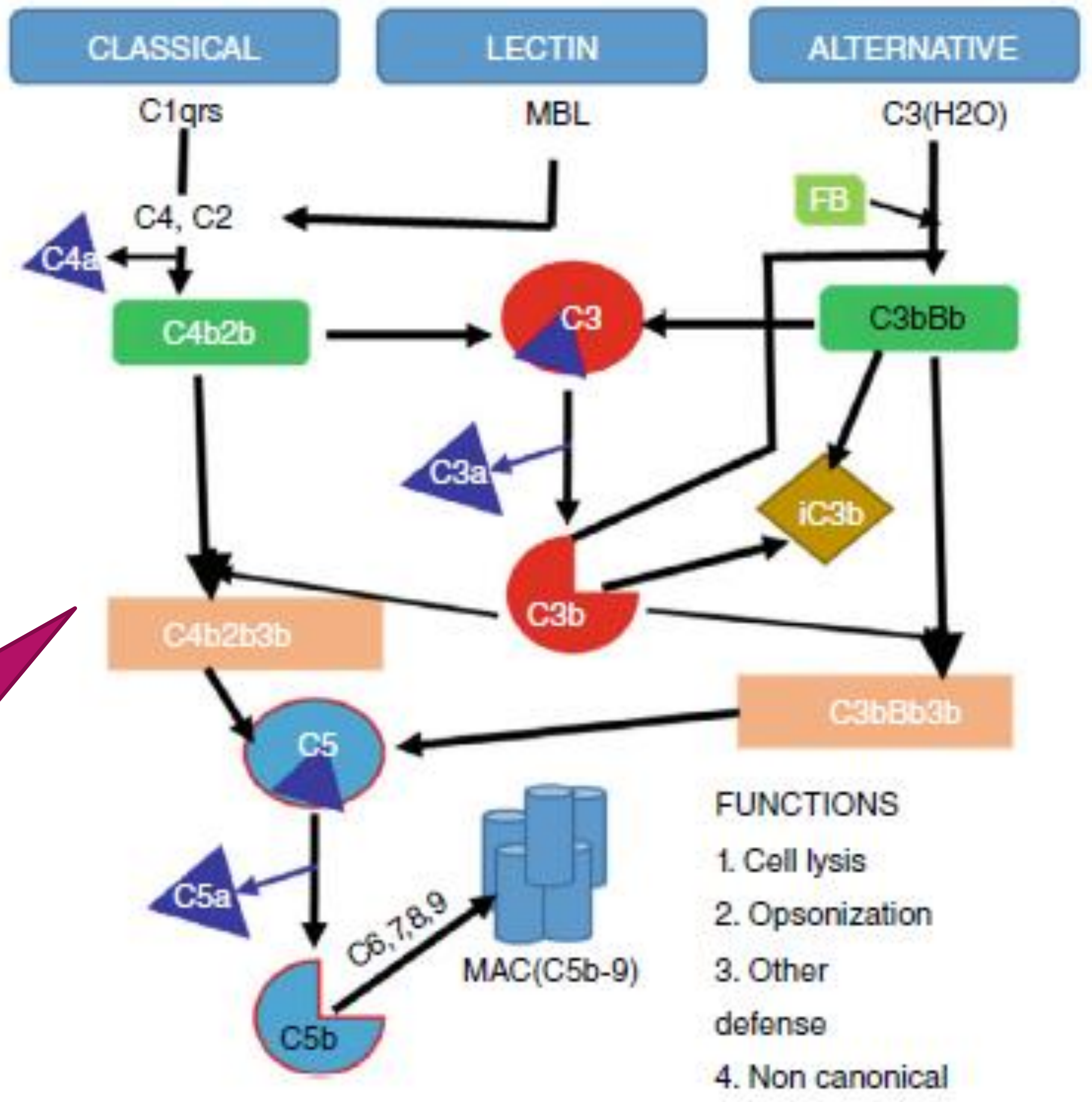


Pathogen Ag



The complement system

- The complement system is made up of approximately 40 proteins of an enzymatic, receptor, and regulatory nature, which circulating in the blood and tissue fluids to participate in a very well-functioning immune system.
- Most of the complement proteins are normally inactive, but in response to the recognition of molecular components of microorganisms they become **sequentially activated** in an **enzyme cascade** the activation of one protein enzymatically cleaves and activates the next protein in the cascade.



Complement can be activated via three different pathways (**Figure**), which can each cause the activation of **C3**, cleaving it into a large fragment, **C3b**, that acts as an **opsonin**, and a small fragment **C3a** (anaphylatoxin) that promotes inflammation. Activated C3 can trigger the **lytic pathway**, which can damage the plasma membranes of cells and some bacteria. C5a, produced by this process, attracts **macrophages** and **neutrophils** and also activates **mast cells**.

Summary: The complement system

The complement system is an enzyme cascade that helps defend against infection via activation of a local inflammatory response. Many complement proteins occur in serum as inactive enzyme precursors (zymogens); others reside on cell surfaces. (See also Overview of the Immune System.)

The complement system bridges innate immunity and acquired immunity by

- Augmenting antibody (Ab) responses and immunologic memory
- Lysing foreign cells
- Clearing immune complexes and apoptotic cells

Complement components have many biologic functions (eg, stimulation of chemotaxis, triggering of mast cell degranulation independent of immunoglobulin E [IgE]).

https://www.youtube.com/watch?v=mLnBqxltTf8

Vaccination and Immune System

BIO-3105, LECTURE-22
12/10/2025

SM RAFIQUL ISLAM, PH.D.

Everyday we faces many things that can make us sick.
These things are called **pathogens**.

Immune system Organs, Tissues and Cells

It can

- **Identify** a threat
- Mount an **attack**
- **Eliminate** the **pathogen**
- Remember the pathogen for **future protection**

But sometimes it goes wrong

- Allergies
- Asthma
- Arthritis

But your **immune system** helps to protect you

Common disorders of the immune system (Overactive/Hyperactive immune system)

It is common for people to have an **over**- or **underactive** immune system. Overactivity of the immune system can take many forms, including:

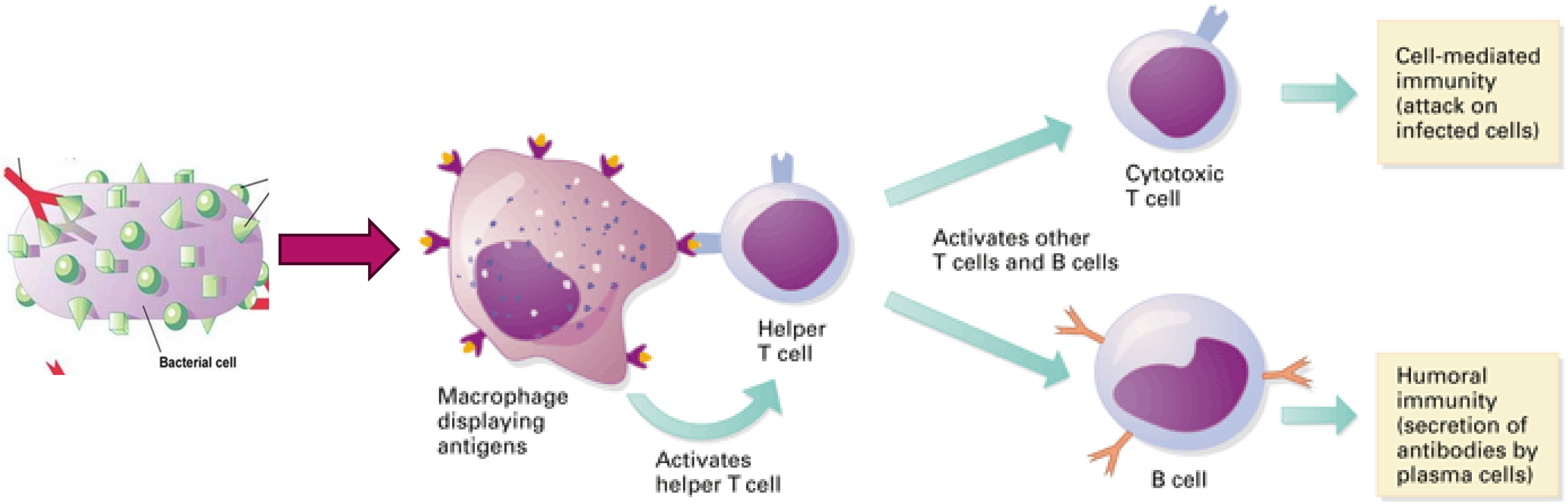
- **Allergic diseases** - where the immune system makes an overly strong response to allergens. Allergic diseases are very common. They include allergies to foods, medications or stinging insects, anaphylaxis (life-threatening allergy), hay fever (allergic rhinitis), sinus disease, asthma, hives (urticaria), dermatitis and eczema
- **Autoimmune diseases** - where the immune system mounts a response against normal components of the body. Autoimmune diseases range from common to rare. They include multiple sclerosis, autoimmune thyroid disease, type 1 diabetes, systemic lupus erythematosus, rheumatoid arthritis and systemic vasculitis.

Common disorders of the immune system (low/poor/weak immune system)

Underactivity of the immune system, also called immunodeficiency, can:

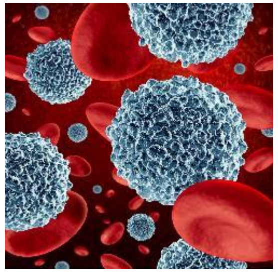
- **be inherited** - examples of these conditions include primary immunodeficiency diseases such as common variable immunodeficiency (CVID), *x-linked severe combined immunodeficiency (SCID)* and complement deficiencies
- arise as a result of **medical treatment** - this can occur due to medications such as corticosteroids or chemotherapy
- **be caused by another disease:** - such as HIV/AIDS or certain types of cancer.
- People who have had an organ transplant need immunosuppression treatment to prevent the body from attacking the transplanted organ.

Cellular response / Immunity



- Once a pathogen (virus, bacteria) invade our body, Macrophages engulf and display the antigens (pathogens)
- This display activates a specific version of helper T (Th) cells activate B cells, cytotoxic T cells, Natural Killer cells and macrophages
- The activated helper T cells in turn stimulate cytotoxic T cells which releases several cytokines and B cells produces antibody.

How does the immune system fight?



White blood cells

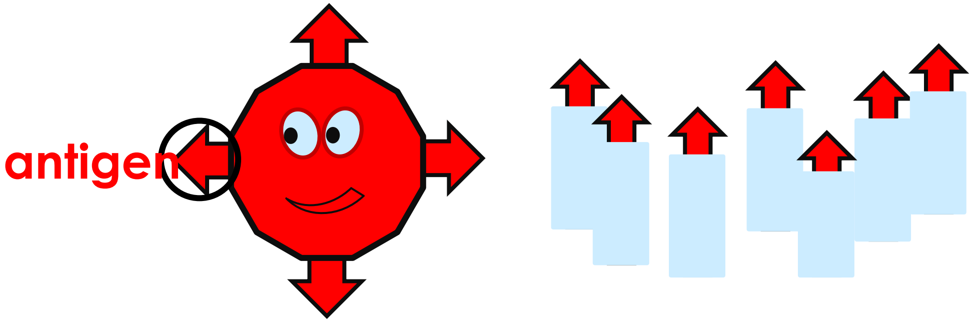
Innate Immune Response

Adaptive Immune Response

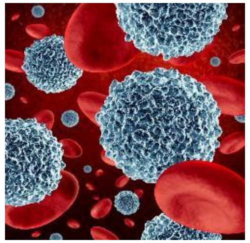
Myeloid cells
(Identifies **My** cells from not **My** cells)

Swallows the **pathogen** whole to kill it and shows **antigens**

Lymphoid cells



How does the immune system fight?



White blood cells

Innate Immune Response

Myeloid cells
(Identifies **My** cells from not **My** cells)

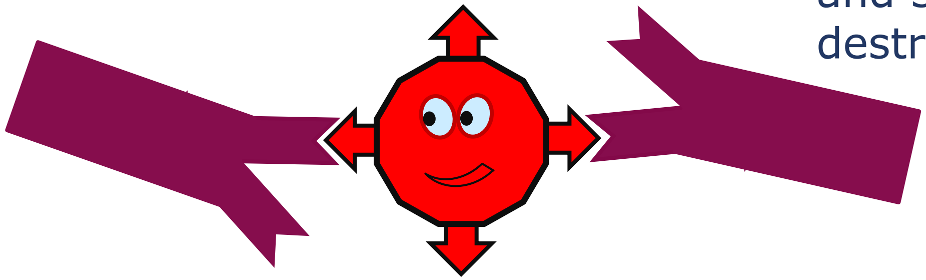
Swallows the **pathogen** whole to kill it and shows **antigens**

Adaptive Immune Response

Lymphoid cells

Makes specific **antibodies** which fit onto specific **pathogen antigens**

and signal for destruction



Immunoglobulin Therapy

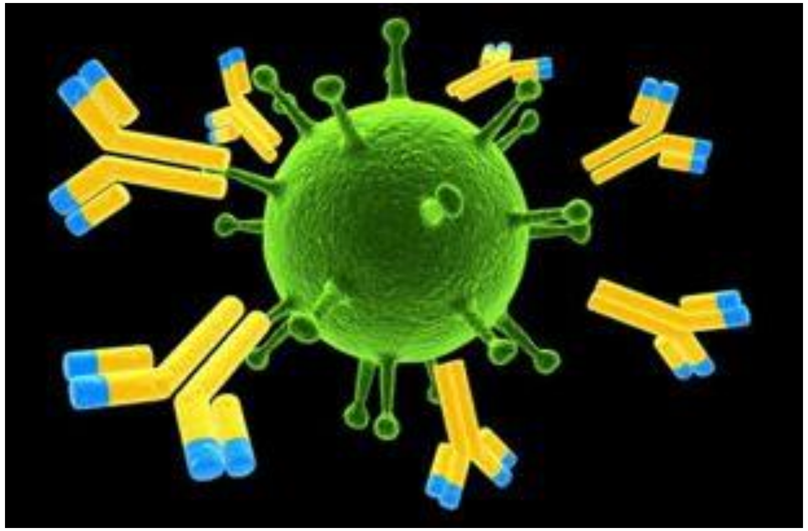
- Immunoglobulins (commonly known as antibodies) are used to treat people who are unable to make enough of their own, or whose antibodies do not work properly. This treatment is known as immunoglobulin therapy.
- Until recently, immunoglobulin therapy in Australia mostly involved delivery of immunoglobulins through a drip into the vein –
- known as intravenous immunoglobulin (IVIg) therapy. Now, subcutaneous immunoglobulin (SCIg) can be delivered into the fatty tissue under the skin, which may offer benefits for some patients. This is known as subcutaneous infusion or SCIg therapy.

Immunoglobulin Therapy

- Subcutaneous immunoglobulin is similar to intravenous immunoglobulin. **It is made from plasma – the liquid part of blood containing important proteins like antibodies.**

Download the SCIg introduction fact sheet to read more about this type of treatment.

- Many health services are now offering SCIg therapy to eligible patients with specific immune conditions. If you are interested, please discuss your particular requirements with your treating specialist.



Immunisation

The immunisations you may need are decided by your health, age, lifestyle and occupation. Together, these factors are referred to as **HALO**, which is defined as:

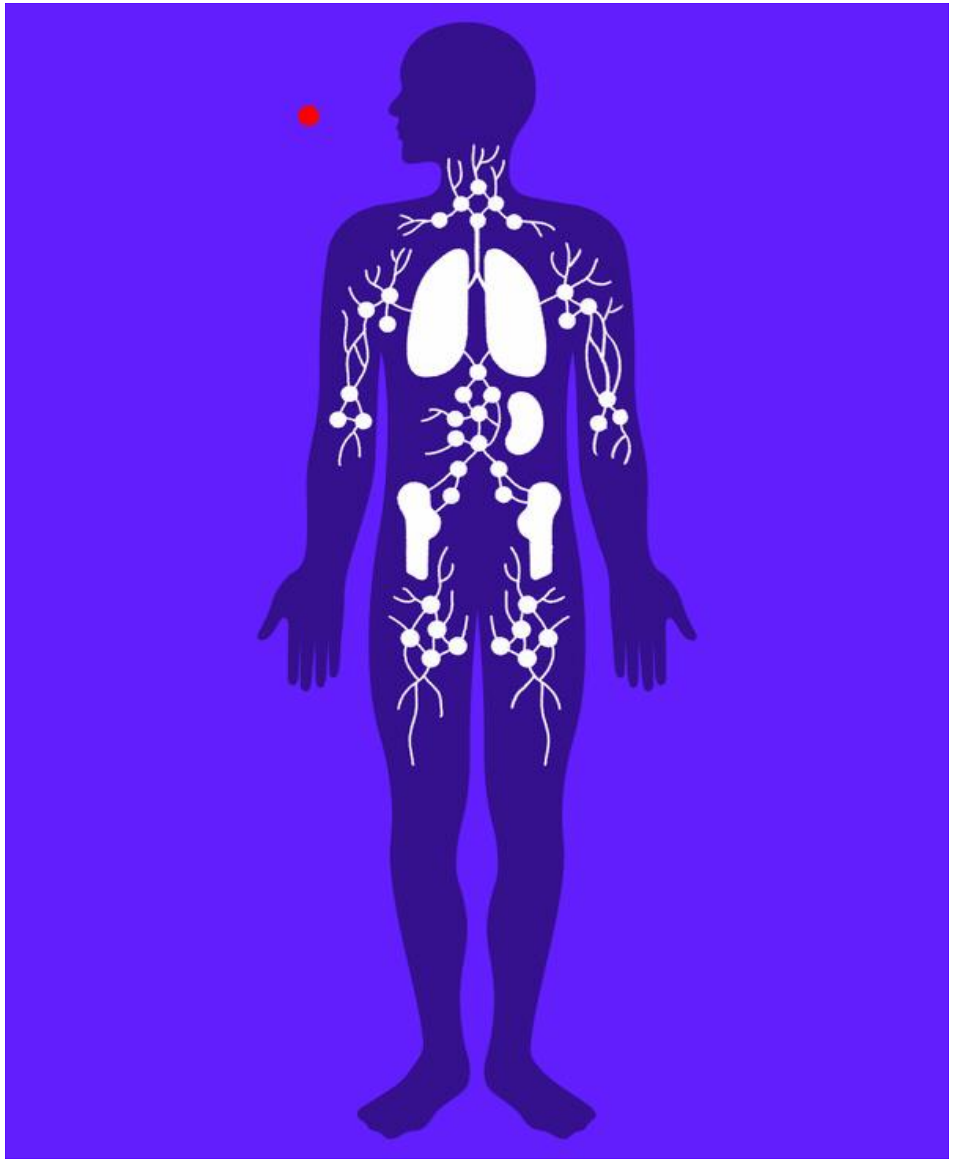
- **health** - some health conditions or factors may make you more vulnerable to vaccine-preventable diseases. For example, premature birth, asthma, diabetes, heart, lung, spleen or kidney conditions, Down syndrome and HIV will mean you may benefit from additional or more frequent immunisations

- **age** - at different ages you need protection from different vaccine-preventable diseases. Australia's National Immunisation Program sets out recommended immunisations for babies, children, older people and other people at risk, such as Aboriginal and Torres Strait Islanders. Most recommended vaccines are available at no cost to these groups; can be an example for us to adopt from for our older people

Immunisation

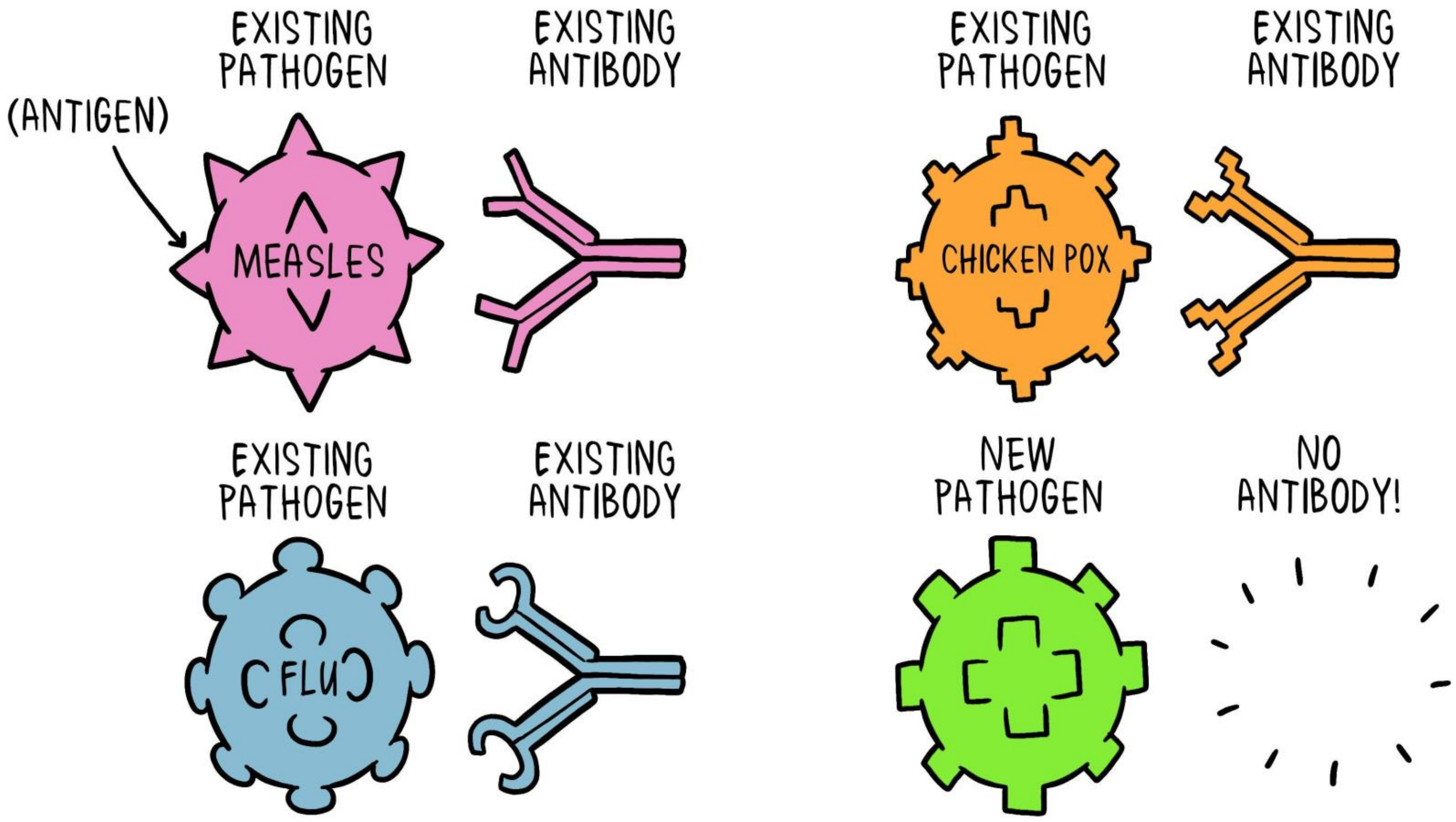
- **lifestyle** - lifestyle choices can have an impact on your immunisation needs. Travelling overseas to certain places, planning a family, sexual activity, smoking, and playing contact sport that may expose you directly to someone else's blood, will mean you may benefit from additional or more frequent immunisations
- **occupation** - you are likely to need extra immunisations, or need to have them more often, if you work in an occupation that exposes you to vaccine-preventable diseases or puts you into contact with people who are more susceptible to problems from vaccine-preventable diseases (such as babies or young children, pregnant women, the elderly, and people with chronic or acute health conditions). For example, if you work in aged care, childcare, healthcare, emergency services or sewerage repair and maintenance, discuss your immunisation needs with your doctor. Some employers help with the cost of relevant vaccinations for their employees.

How do vaccines work?



- Germs are all around us, both in our environment and in our bodies. When a person is susceptible and they encounter a harmful organism, it can lead to disease and death.
- The body has many ways of defending itself against **pathogens** (disease-causing organisms). Skin, mucus, and cilia (microscopic hairs that move debris away from the lungs) all work as physical barriers to prevent pathogens from entering the body in the first place.
- When a pathogen does infect the body, our body's defences, called the immune system, are triggered and the pathogen is attacked and destroyed or overcome.

How do vaccines work?

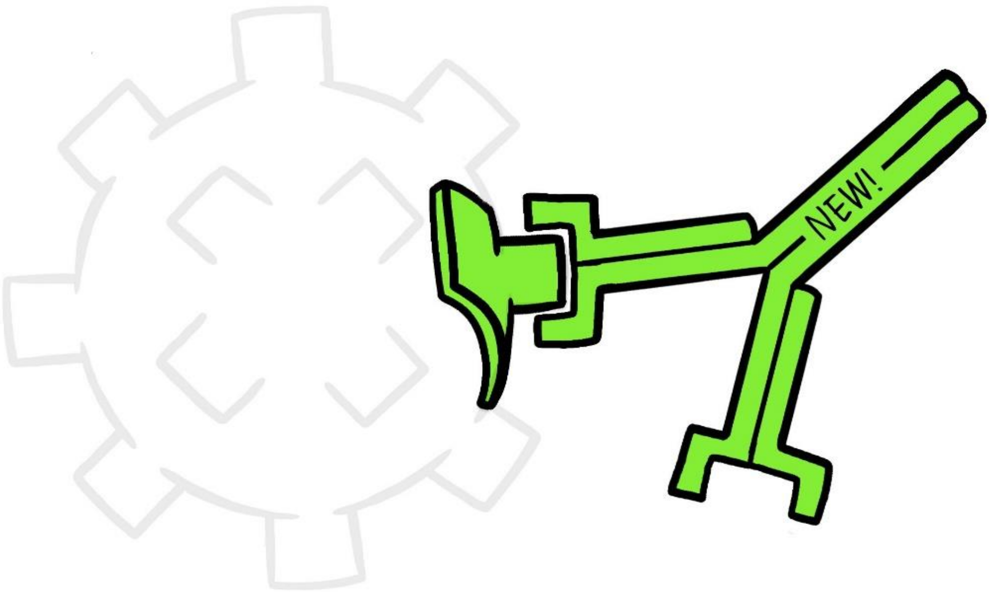


When a new pathogen or disease enters our body, it introduces a new antigen. For every new antigen, our body needs to build a specific antibody that can grab onto the antigen and defeat the pathogen.

How do vaccines work?

VACCINE

NEW ANTIBODY



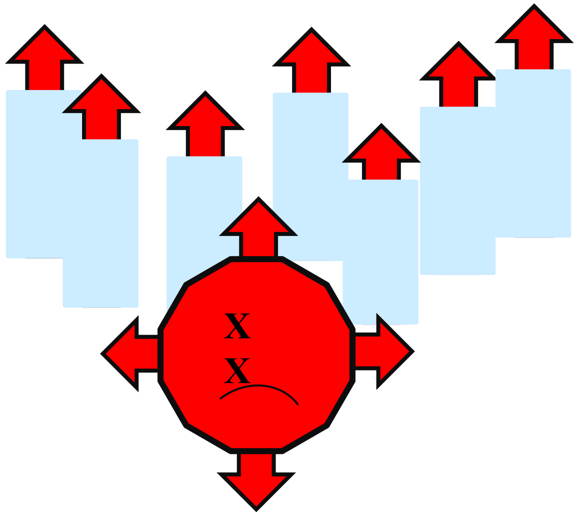
A VACCINE is a tiny weakened non-dangerous fragment of the organism and includes parts of the antigen. It's enough that our body can learn to build the specific antibody. Then if the body encounters the real antigen later, as part of the real organism, it already knows how to defeat it.

Vaccines, How does it work?

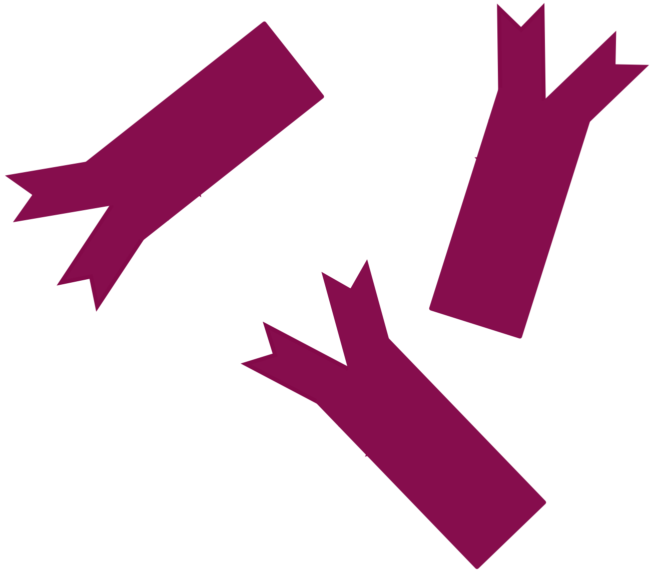
The 3 types of **vaccines** contain

- A weakened **pathogen** (not strong enough to make us sick) → **live-attenuated vaccine**
- A killed **pathogen** (can't make us sick) → **inactivated vaccine**
- A small part of a **pathogen** (like an **antigen**) → **subunit vaccine**

↓



All vaccines contain **antigens** which trigger the **immune system** to create specific **antibodies**



↓

A signal for **pathogen** destruction in future

How do vaccines work?

How vaccines help

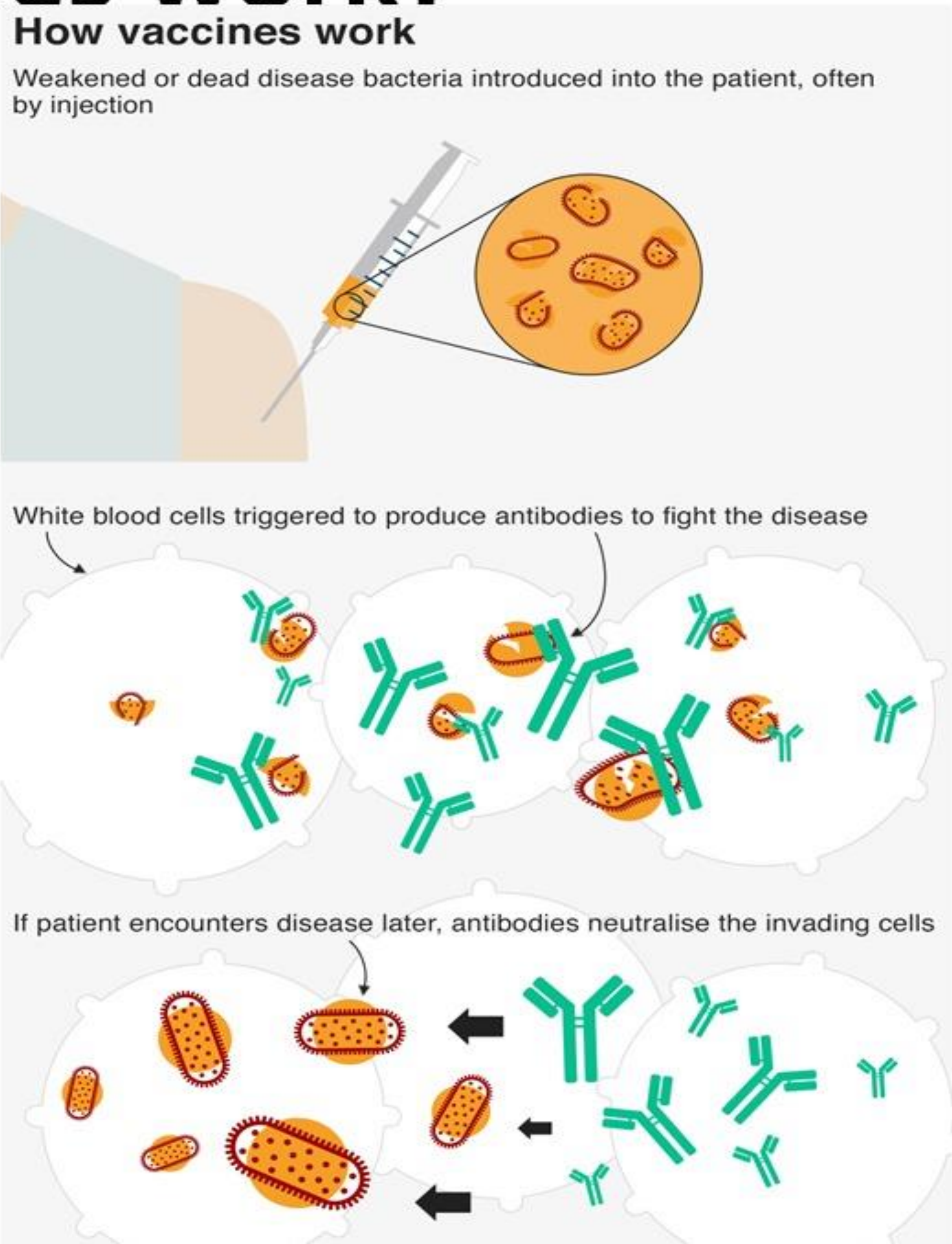
- Vaccines contain weakened or inactive parts of a particular organism (antigen) that triggers an immune response within the body.
- Newer vaccines contain the blueprint for producing antigens rather than the antigen itself.
- Regardless of whether the vaccine is made up of the antigen itself or the blueprint so that the body will produce the antigen, this weakened version will not cause the disease in the person receiving the vaccine, but it will prompt their immune system to respond much as it would have on its first reaction to the actual pathogen.

- Some vaccines require multiple doses, given weeks or months apart.
- This is sometimes needed to allow for the production of long-lived antibodies and development of memory cells.

- In this way, the body is trained to fight the specific disease-causing organism, building up memory of the pathogen so as to rapidly fight it if and when exposed in the future.

How do vaccines work?

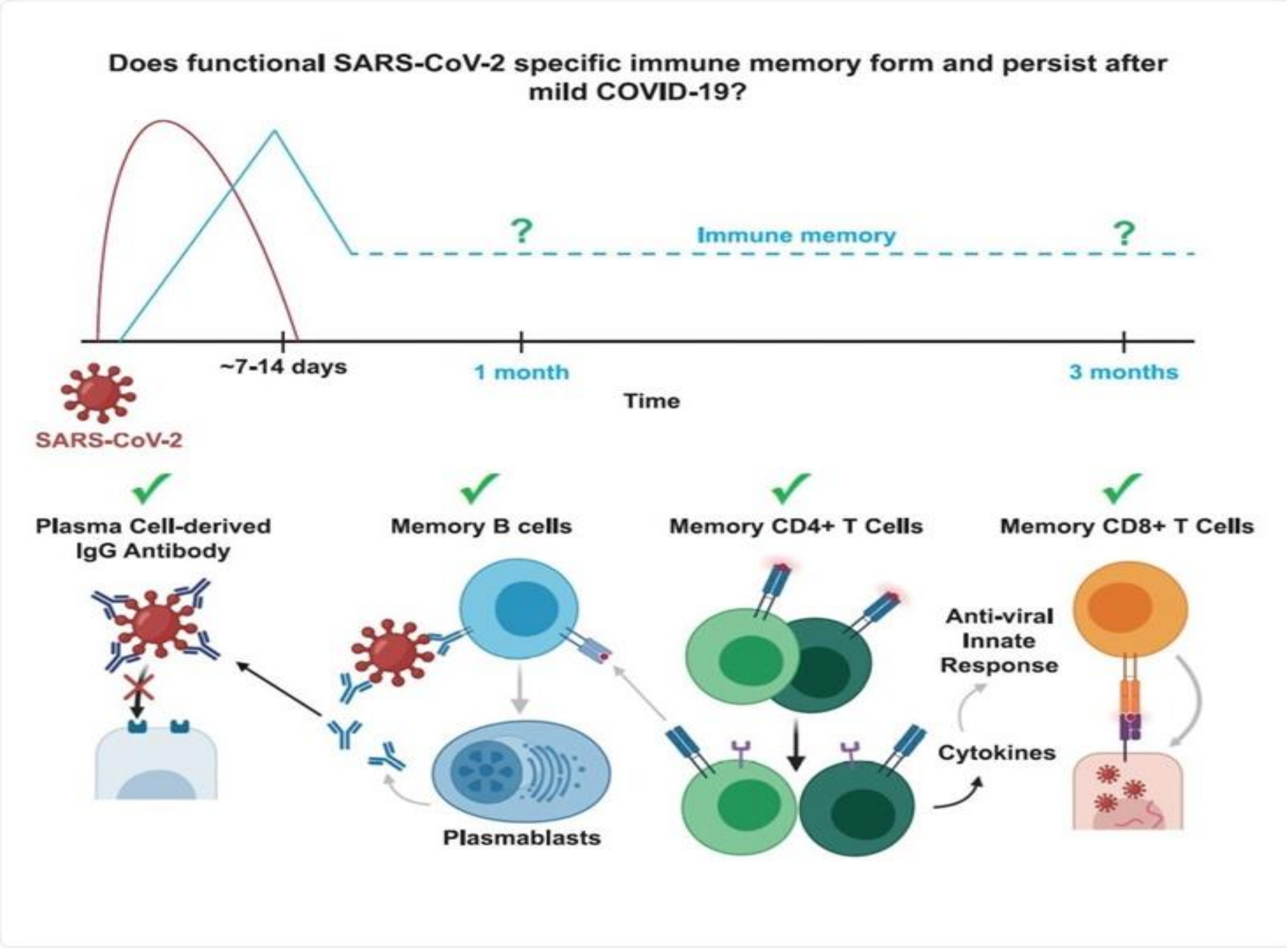
- In the meantime, the person is susceptible to becoming ill.
- Once the antigen-specific antibodies are produced, they work with the rest of the immune system to destroy the pathogen and stop the disease.
- Antibodies to one pathogen generally don't protect against another pathogen except when two pathogens are very similar to each other, like cousins.
- Once the body produces antibodies in its primary response to an antigen, it also creates antibody-producing memory cells, which remain alive even after the pathogen is defeated by the antibodies. I
- f the body is exposed to the same pathogen more than once, the antibody response is much faster and more effective than the first time around because the memory cells are at the ready to pump out antibodies against that antigen.
- This means that if the person is exposed to the dangerous pathogen in the future, their immune system will be able to respond immediately, protecting against disease.



How do vaccines work?

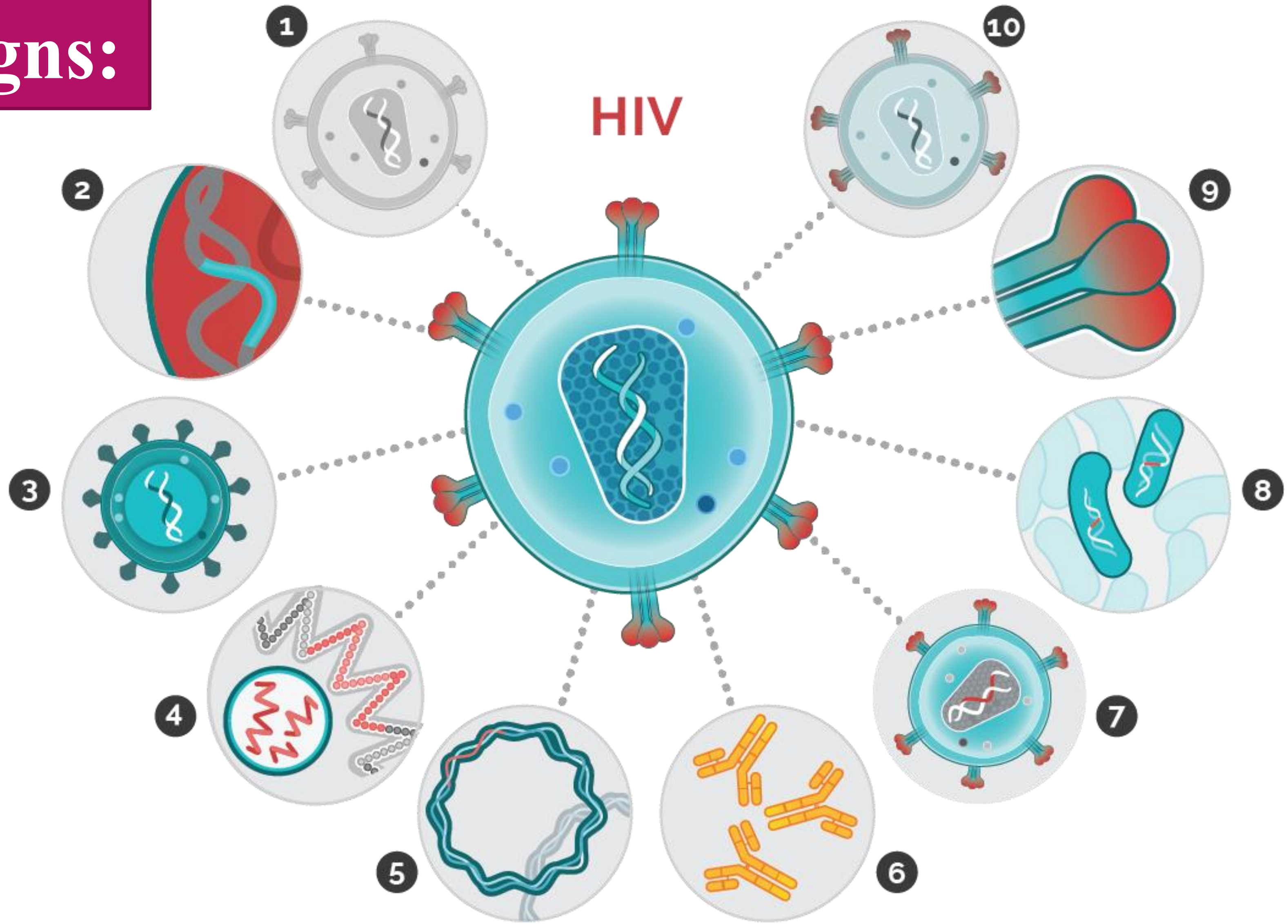
The body's natural response

- A pathogen is a bacterium, virus, parasite or fungus that can cause disease within the body. Each pathogen is made up of several subparts, usually unique to that specific pathogen and the disease it causes. The subpart of a pathogen that causes the formation of antibodies is called an antigen. The antibodies produced in response to the pathogen's antigen are an important part of the immune system. You can consider antibodies as the soldiers in your body's defense system. Each antibody, or soldier, in our system is trained to recognize one specific antigen. We have thousands of different antibodies in our bodies. When the human body is exposed to an antigen for the first time, it takes time for the immune system to respond and produce antibodies specific to that antigen.



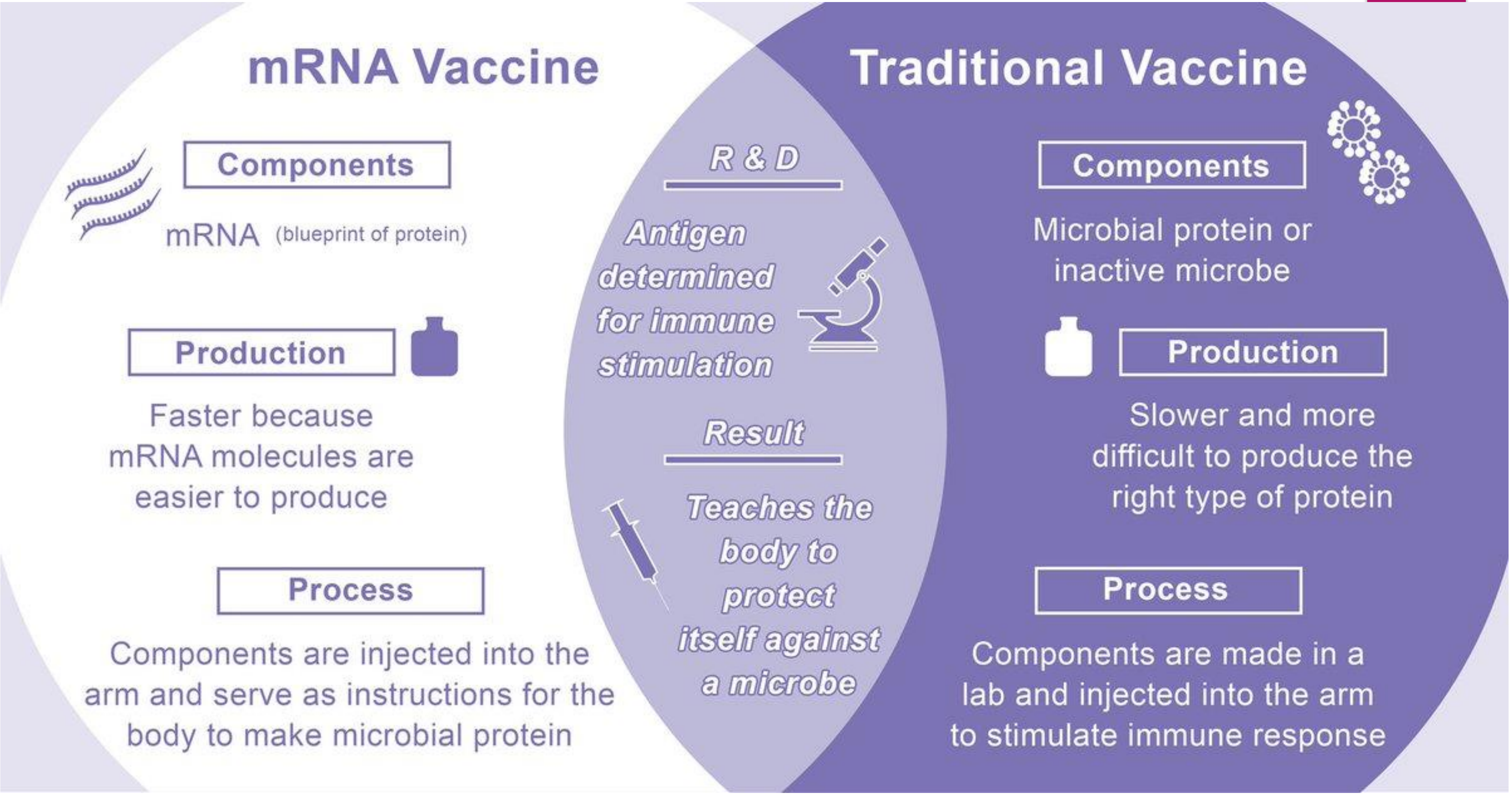
Vaccine Designs:

Choice of antigen to induce antibody responses



1) Whole-inactivated / inactivated HIV; 2) synthetic peptide / laboratory-made piece of protein; 3) recombinant viral vector / another virus carries pieces of HIV; 4) mRNA / mRNA carries pieces of HIV; 5) DNA / DNA carries pieces of HIV; 6) broadly neutralizing antibodies / binds to and neutralizes HIV; 7) virus-like particles / same shape as HIV, insides changed; 8) recombinant bacterial vector / bacteria used to carry pieces of HIV; 9) recombinant sub-unit / HIV protein made in a lab; 10) live-attenuated / weakened HIV

Advantage of mRNA Vaccines



Types of Vaccines

- **mRNA vaccines** contain material from the virus that causes COVID-19 that gives our cells instructions for how to make a harmless protein that is unique to the virus. After our cells make copies of the protein, they destroy the genetic material from the vaccine. Our bodies recognize that the protein should not be there and build T-lymphocytes and B-lymphocytes that will remember how to fight the virus that causes COVID-19 if we are infected in the future.
- **Protein subunit vaccines** include harmless pieces (proteins) of the virus that causes COVID-19 instead of the entire germ. Once vaccinated, our bodies recognize that the protein should not be there and build T-lymphocytes and antibodies that will remember how to fight the virus that causes COVID-19 if we are infected in the future.
- **Vector vaccines** contain a modified version of a different virus than the one that causes COVID-19. Inside the shell of the modified virus, there is material from the virus that causes COVID-19. This is called a "viral vector." Once the viral vector is inside our cells, the genetic material gives cells instructions to make a protein that is unique to the virus that causes COVID-19. Using these instructions, our cells make copies of the protein. This prompts our bodies to build T-lymphocytes and B-lymphocytes that will remember how to fight that virus if we are infected in the future.

Vaccines

Benefits

- ✓ Provide protection from **pathogens** which can kill or make you sick
- ✓ Control and eradication of once common **pathogens** like polio and smallpox
- ✓ Stops outbreaks and **global epidemics**

Risks/challenges

No medicine is 100% effective – people can sometimes still catch the pathogen but this is very rare

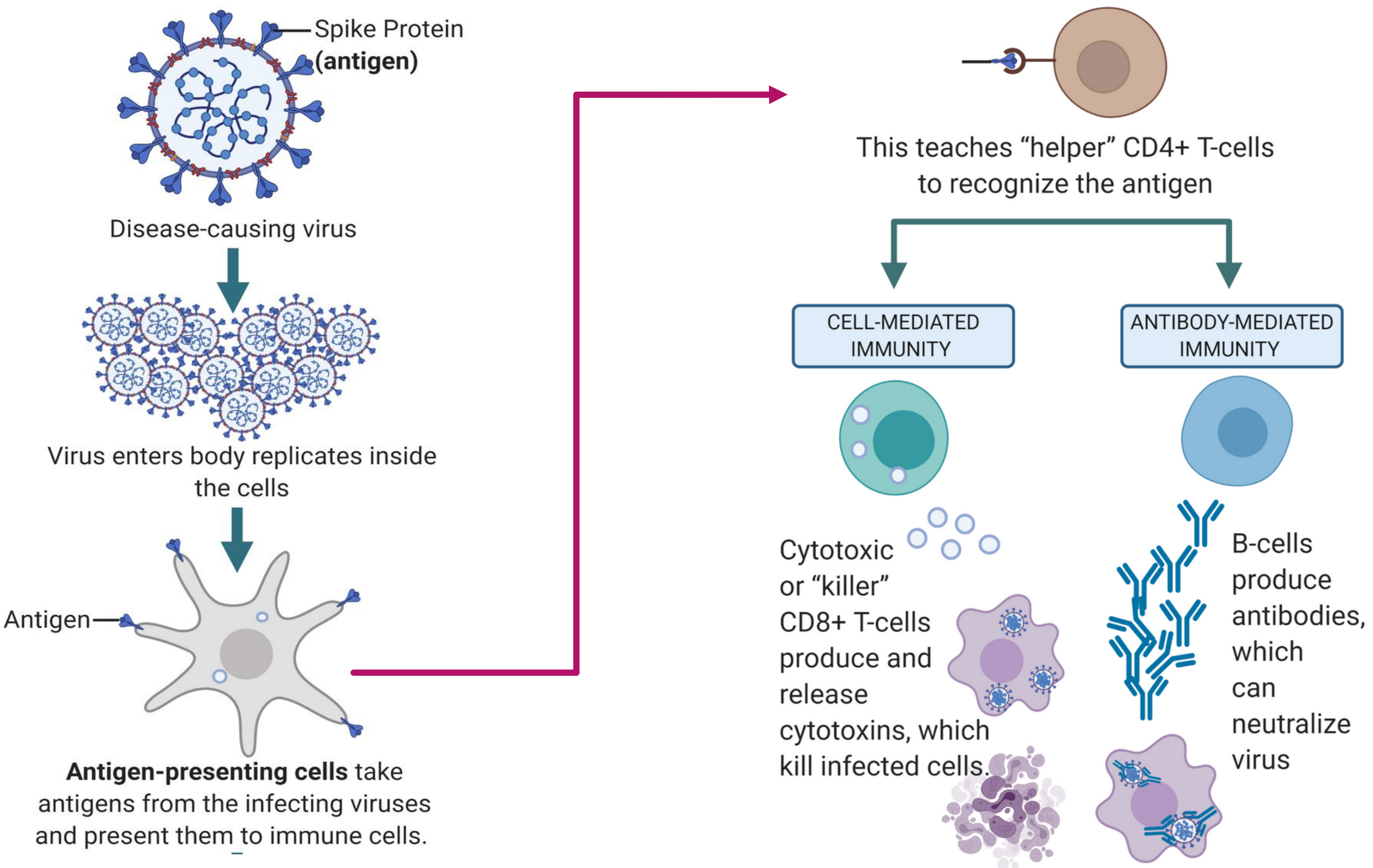
Side effects like swelling and fevers

Not all pathogens have a vaccine for example malaria parasites

How COVID-19 Vaccines Work

- Different types of vaccines work in different ways to offer protection.
- But with all types of vaccines, the body is left with a supply of "memory" T-lymphocytes as well as B-lymphocytes that will remember how to fight that virus in the future.
- It typically takes a few weeks after vaccination for the body to produce T-lymphocytes and B-lymphocytes. Therefore, it is possible that a person could be infected with the virus that causes COVID-19 just before or just after vaccination and then get sick because the vaccine did not have enough time to provide protection.
- Sometimes after vaccination, the process of building immunity can cause symptoms, such as fever. These symptoms are normal and are signs that the body is building immunity.

Summary-1
Immune responses against a Virus/Antigen/Vaccine



Summary-2
Immune responses against a Virus/Antigen/Vaccine

Pathogens

- E.coli
- Athlete's foot
- Malaria
- Flu (Influenza)

- Make you sick and some people can die

- Can spread from person to person

Immune system

- Helps to protect you from **pathogens**

- **Innate immune system** is fast but has no memory

- **Adaptive immune system** is slower by remembers
  Produces **antibodies**

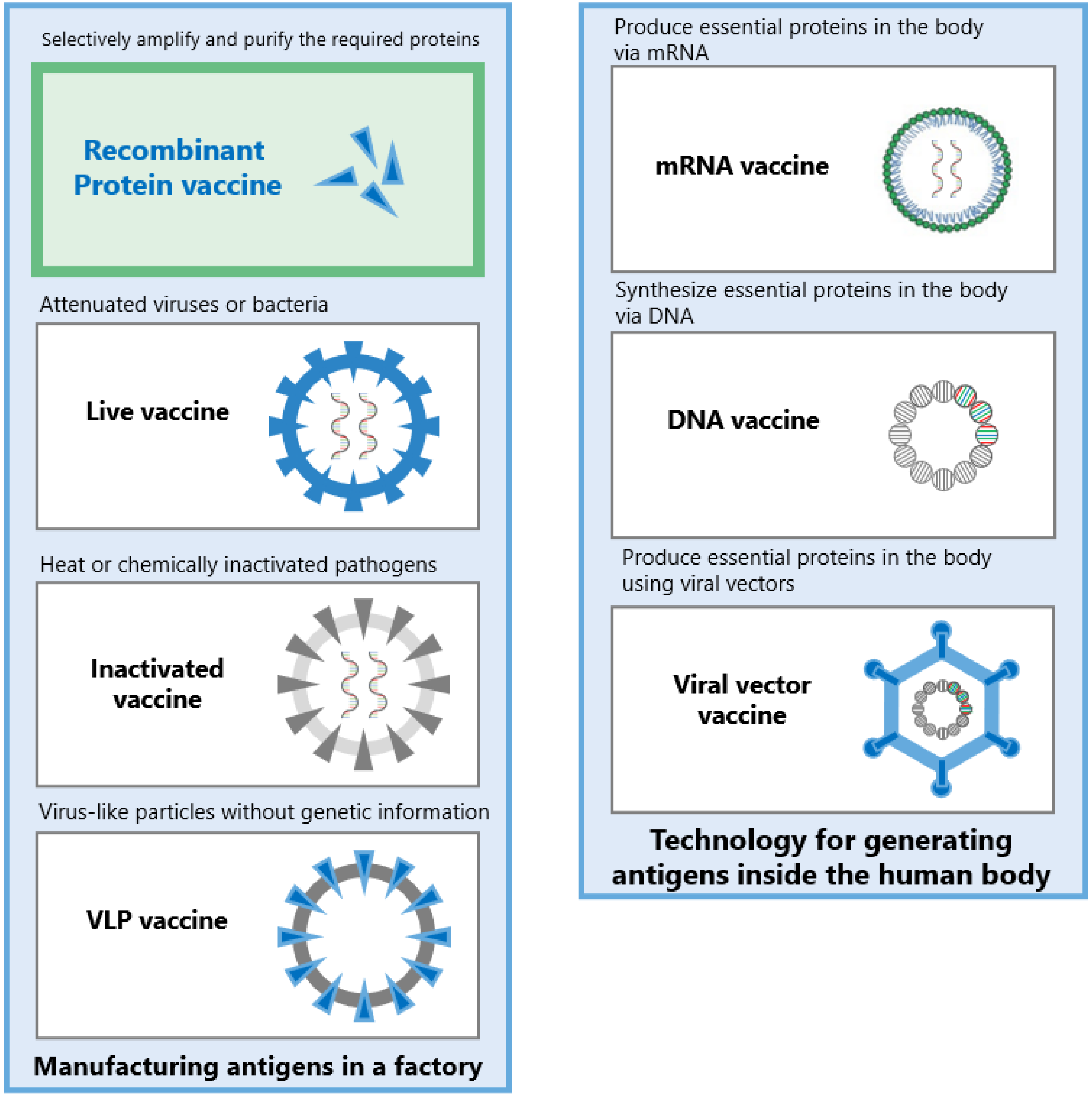
Vaccination

- Exposes the **immune system** to **pathogens** in a **safe** way

- Protects you from **pathogens** by generating **antibodies**

- **Herd immunity** protects people that can't be **vaccinated**
    - But the **vaccinated** population must be high

Further Reading


Selectively amplify and purify the required proteins
Recombinant Protein vaccine
Attenuated viruses or bacteria
Live vaccine
Heat or chemically inactivated pathogens
Inactivated vaccine
Virus-like particles without genetic information
VLP vaccine
Manufacturing antigens in a factory
Produce essential proteins in the body via mRNA
mRNA vaccine
Synthesize essential proteins in the body via DNA
DNA vaccine
Produce essential proteins in the body using viral vectors
Viral vector vaccine
Technology for generating antigens inside the human body

The immunological mechanisms of inflammation - local

- Activation of haemocoagulation → synthesis of kinins (e.g.bradykinin) → vasodilation, ↑vascular permeability → oedema, pain
- Cell mediated response – acute (neutrophils), chronic (macrophages, lymphocytes) → tumor etc.
- Healing – restoration of tissue architecture, scare tissue development

The immunological mechanisms of inflammation - systemic

- Fever (IL-1, TNF)
- Leucocytosis (IL-1)
- Production of acute phase proteins
- Complement system activation
- Specific response (production of antigen-specific antibodies and T cells)

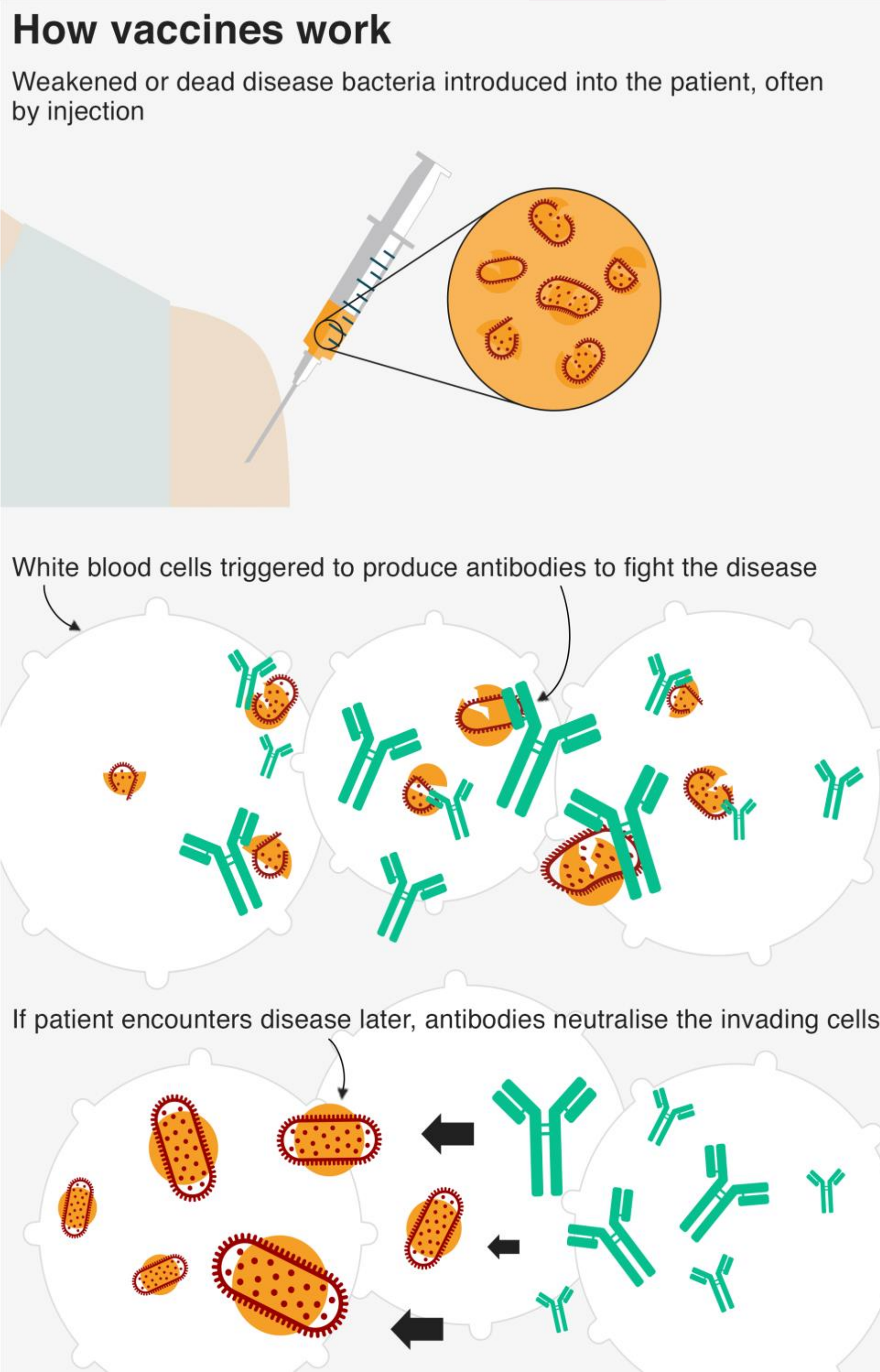
How do vaccines work?

Herd immunity

- When someone is vaccinated, they are very likely to be protected against the targeted disease. But not everyone can be vaccinated. People with underlying health conditions that weaken their immune systems (such as cancer or HIV) or who have severe allergies to some vaccine components may not be able to get vaccinated with certain vaccines. These people can still be protected if they live in and amongst others who are vaccinated.
- When a lot of people in a community are vaccinated the pathogen has a hard time circulating because most of the people it encounters are immune. So the more that others are vaccinated, the less likely people who are unable to be protected by vaccines are at risk of even being exposed to the harmful pathogens. This is called herd immunity.
- This is especially important for those people who not only can't be vaccinated but may be more susceptible to the diseases we vaccinate against.
- No single vaccine provides 100% protection, and herd immunity does not provide full protection to those who cannot safely be vaccinated. But with herd immunity, these people will have substantial protection, thanks to those around them being vaccinated.
- Vaccinating not only protects yourself, but also protects those in the community who are unable to be vaccinated. If you are able to, get vaccinated.

How do vaccines work?

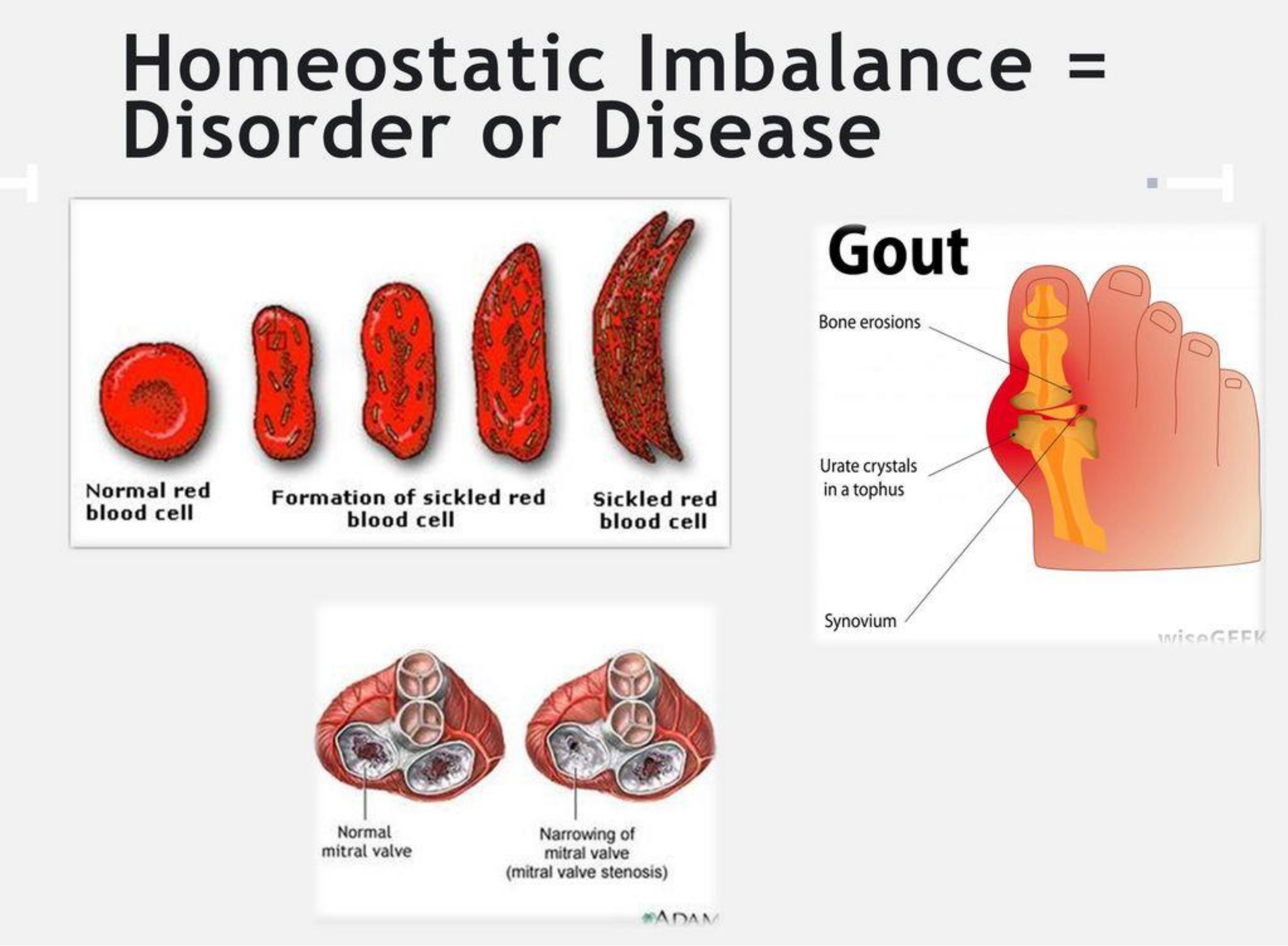
- In the meantime, the person is susceptible to becoming ill.
- Once the antigen-specific antibodies are produced, they work with the rest of the immune system to destroy the pathogen and stop the disease.
- Antibodies to one pathogen generally don't protect against another pathogen except when two pathogens are very similar to each other, like cousins.
- Once the body produces antibodies in its primary response to an antigen, it also creates antibody-producing memory cells, which remain alive even after the pathogen is defeated by the antibodies. I
- f the body is exposed to the same pathogen more than once, the antibody response is much faster and more effective than the first time around because the memory cells are at the ready to pump out antibodies against that antigen.
- This means that if the person is exposed to the dangerous pathogen in the future, their immune system will be able to respond immediately, protecting against disease.



Disease as Homeostatic Imbalance

Diabetes: A Disease of Failed Homeostasis

- Diabetes, a metabolic disorder caused by excess blood glucose levels, is a key example of disease caused by failed homeostasis. In ideal circumstances, homeostatic control mechanisms should prevent this imbalance from occurring. However, in some people, the mechanisms do not work efficiently enough or the amount of blood glucose is too great to be effectively managed. In these cases, medical intervention is necessary to restore homeostasis and prevent permanent organ damage.



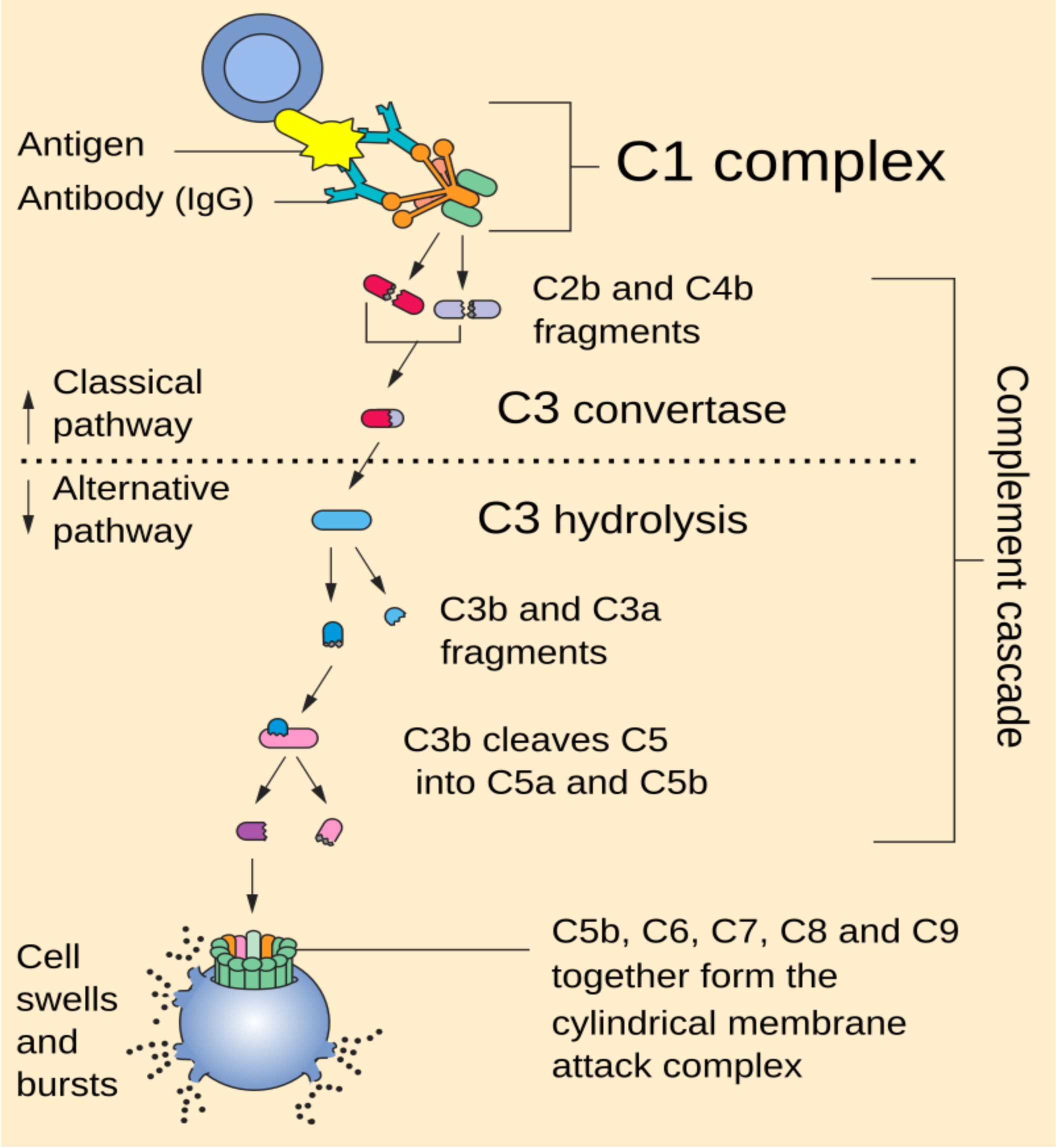
Natural killer cells (NK cells)

- instead of attacking the foreign invaders, NK cells attack the body's own cells that have become infected by viruses

- they also attack potential cancer cells, often before they form tumors

- they bind to cells using an antibody "bridge", then kill it by secreting a chemical (perforin) that makes holes in the cell membrane of the target cell (ie. Bacteria) and die.

Mechanism of complement cascade activation and neutralization of the foreign pathogens



Complement affects different organs:
Dysfunctional complement can affect one or all organs